

General Comments

Chapter 12 addresses the contemporary energy-related systems found in the code and serves as an introduction to a wide range of systems to generate and store energy in, on and adjacent to buildings and facilities. The expansion of such energy systems is related to meeting today's energy and environmental demands. Part of this demand is the desire for energy independence and stability. Appropriate criteria to address the safety of such systems in building and fire codes are an important part of protecting the public at large, building occupants and emergency responders. Prior to the 2018 code, requirements related to energy systems such as standby power systems, photovoltaic (PV) systems and stationary battery systems were scattered in various locations in Chapter 6, which addresses building services and systems. However, with the expansion of the requirements to more general energy storage systems and stationary fuel cell systems, a chapter dedicated to such issues was necessary. This chapter provides an appropriate location for the addition of future energy systems.

The provisions are broken into several main sections as follows:

- Section 1203. Emergency and standby power systems. This section outlines all of the systems which require standby or emergency power. In addition, this section provides the necessary requirements surrounding the appropriate installation, testing and maintenance of such systems. These provisions mirror those in Chapter 27 of the *International Building Code*® (IBC®).
- Section 1204. These provisions address the safe operation of portable generators in any application. The focus is on proper listings, use, grounding and locations.
- Section 1205. Solar photovoltaic power systems. These provisions address the needs of the fire department fighting a fire with PVs installed. The provisions focus on both providing pathways and proper signage to understand where and how the system can be shut down. There is also a general reference to the *National Electrical Code*® (NEC®) (NFPA 70) for all other applicable requirements.
- Section 1206. Stationary fuel cell power systems. These systems are becoming more prevalent in buildings and the hazards are being addressed. They often use hydrogen gas and cng as a fuel, which can pose a significant hazard if not properly managed.
- Section 1207. Energy storage systems include a wide variety of battery technologies that are used in various applications. This includes standalone installations and systems within occupied buildings. This section also addresses mobile electrical energy storage systems (ESS). These types of systems are increasing as the technology develops and without proper regulation, they pose a threat to building occupants and emergency responders.

Purpose

The purpose of Chapter 12 is to centralize energy-related issues regulated by the code to properly manage potential hazards to occupants and firefighters. Additionally, it provides the framework to address new and emerging energy systems and the related risks. For example, the battery storage system section (Section 1207) specifically has provisions for other battery technologies not yet recognized. New systems can be addressed through hazard analysis and testing.

SECTION 1201—GENERAL

1201.1 Scope. The provisions of this chapter shall apply to the installation, operation, maintenance, repair, retrofitting, testing, commissioning and decommissioning of energy systems used for generating or storing energy, including but not limited to energy storage systems under the exclusive control of an electric utility or lawfully designated agency. It shall not apply to equipment associated with the generation, control, transformation, transmission, or distribution of energy installations that is under the exclusive control of an electric utility or lawfully designated agency. Energy storage systems regulated by Section 1207 shall comply with this chapter, as appropriate, and NFPA 855.

C The focus of this chapter is on energy systems as they pertain to energy generation and storage. In the case of photovoltaic systems, much of the requirements are focused on the safety of firefighters who may need to access a roof or portions of the building containing such systems.

This section specifically notes that the chapter is not focused on the distribution and transmission that is controlled and regulated by an electric utility. However, energy storage systems would be regulated by this chapter regardless of who owns and operates such systems. The focus is on the systems as they affect occupants and firefighter safety along with limiting the size of an event. This section also references NFPA 855. NFPA 855 is closely aligned with the requirements of Section 1207. NFPA 855 contains expanded requirements and a large amount of guidance material.

1201.2 Electrical wiring and equipment. Electrical wiring and equipment used in connection with energy systems shall be installed and maintained in accordance with this chapter, Section 603 and NFPA 70.

C The intention is that in addition to requirements found in this chapter, anything electrical in nature is required to comply with the NEC. This is the primary referenced standard throughout the International Codes® (I-Codes®) for all electrical issues.

1201.3 Mixed system installation. Where mixed systems are *approved*, the aggregate nameplate kWh energy of all energy storage systems in a *fire area* shall not exceed the maximum quantity specified for any of the energy systems in this chapter. Where required by the *fire code official*, a hazard mitigation analysis shall be provided and *approved* in accordance with Section 104.2.2 to evaluate any potential adverse interaction between the various energy systems and technologies.

C The concept of on-site energy storage and generation within or supporting a specific building is becoming more prevalent. The methods and types of installations are often mixed and must be looked at in terms of their overall fire risk. In some cases, this chapter already addresses this issue. For example, in Section 1207 there are specific overall kWh limitations where different battery storage systems are used in a single space without proper separation. However, the fact that a fuel cell may be located in a fire area with an ESS would need to be addressed. This section empowers the fire code official to require evaluation of the impact of such systems.

SECTION 1202—DEFINITIONS

1202.1 Definitions. The following terms are defined in Chapter 2:

BATTERY SYSTEM, STATIONARY STORAGE.

BATTERY TYPES.

CAPACITOR ENERGY STORAGE SYSTEM.

CRITICAL CIRCUIT.

EMERGENCY POWER SYSTEM.

ENERGY STORAGE MANAGEMENT SYSTEMS.

ENERGY STORAGE SYSTEM (ESS).

ENERGY STORAGE SYSTEM, ELECTROCHEMICAL.

ENERGY STORAGE SYSTEM, MOBILE.

ENERGY STORAGE SYSTEM, WALK-IN UNIT.

ENERGY STORAGE SYSTEM CABINET.

ENERGY STORAGE SYSTEM COMMISSIONING.

ENERGY STORAGE SYSTEM DECOMMISSIONING.

FUEL CELL POWER SYSTEM, STATIONARY.

PORTABLE GENERATOR.

STANDBY POWER SYSTEM.

C This section lists terms that are specifically associated with the subject matter of this chapter. It is important to emphasize that these terms are not exclusively related to this chapter but may or may not also be applicable where the term is used elsewhere in the code.

Definitions of terms can help in the understanding and application of the code requirements. The purpose for including a list within this chapter is to provide more convenient access to terms that may have a specific or limited application within this chapter. For the complete definition and associated commentary, refer back to Chapter 2. Terms that are italicized provide a visual identification throughout the code that a definition exists for that term. The use and application of all defined terms are set forth in Section 201.

SECTION 1203—EMERGENCY AND STANDBY POWER SYSTEMS

1203.1 General. Emergency power systems and standby power systems required by this code or the *International Building Code* shall comply with Sections 1203.1.1 through 1203.1.9.

C This section states that all required emergency and standby power systems should comply with various subsections. These subsections provide more detailed direction on required installation for such systems. These provisions are consistent with the standards referenced by the code and provide basic information to the code official.

1203.1.1 Stationary generators. Stationary emergency and standby power generators required by this code shall be *listed* in accordance with UL 2200.

C UL 2200 was developed at the request of building and fire officials to provide a standard to evaluate the safety and reliability of stationary engine generators. UL 2200 establishes a basis for this evaluation. As noted in previous sections, all electri-

cal systems and appliances must comply with NFPA 70; however, this is a specific listing that demonstrates that a specific generator meets the necessary functional and safety requirements

This section applies only to generator sets that are chosen and permanently installed to provide an emergency or standby power supply for the required, fixed building facilities indicated in Section 1203.2. UL 2200 defines a stationary unit as “An engine generator that is intended to be hard wired and/or permanently installed.” The section would not apply to mobile units such as truck-, trailer- or skid-mounted or other nonfixed, nonpermanent units. Such nonpermanently installed units would not be approved for purposes of complying with the requirements of Section 1203. Similarly, stationary generator sets that are installed for purposes other than to supply emergency or standby power required by the code would not be required to be listed to UL 2200.

1203.1.2 Fuel line piping protection. Fuel lines supplying a generator set inside a high-rise building shall be separated from areas of the building other than the room the generator is located in by one of the following methods:

1. A fire-resistant pipe-protection system that has been tested in accordance with UL 1489. The system shall be installed as tested and in accordance with the manufacturer’s installation instructions, and shall have a rating of not less than 2 hours. Where the building is protected throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1, the required rating shall be reduced to 1 hour.
2. An assembly that has a *fire-resistance rating* of not less than 2 hours. Where the building is protected throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1, the required *fire-resistance rating* shall be reduced to 1 hour.
3. Other *approved* methods.

C This section is intended to require fuel lines supplying a generator set inside a building to be separated with fire-resistance-rated construction from areas of the building other than the room in which the generator is located.

These provisions extend the requirement to any building that has a generator that is separated from the rest of the building. It is common for diesel-fueled generators to have a day tank and resupply the day tank via remote fuel oil tanks. The fuel line piping from those remote tanks to the generator can be exposed to the same fire incident that the generator has been protected against. Loss of the fuel line due to fire exposure has the same impact as loss of the generator itself.

The language only refers to “fuel lines” to also provide protection in those cases where a gaseous fuel supply is approved for use.

This section offers several options, including the use of a UL 1489 fire-resistant piping system. This provides fire resistance to the pipe itself versus being located within the building construction. Generally, this section is intended to provide 2-hour protection in a building without automatic sprinkler protection, complying with NFPA 13. Where an NFPA 13 system is provided, the fire resistance can be reduced to 1 hour.

1203.1.3 Installation. Emergency power systems and standby power systems shall be installed in accordance with the *International Building Code*, NFPA 70, NFPA 110 and NFPA 111.

C Emergency power systems are intended to provide electrical power for life safety systems, such as egress illumination, emergency communications and processes involving the handling and use of hazardous materials. In other words, emergency power is required where the loss of normal power would endanger occupants. Such systems are covered in Article 700 of NFPA 70 and one of their key features is the required response time of 10 seconds or less. The time between loss of normal power and the provision of emergency power must be kept short to prevent putting occupants at risk. This is especially important during an emergency event, such as a building fire, but is also important to prevent a situation that reduces the ability for occupants to properly navigate the building. Without proper illumination, this can affect the ability of occupants to safely relocate or egress the building.

Standby power systems are covered in Article 701 of NFPA 70 and are intended to provide electrical power for loads not as critical in terms of transfer time as those requiring emergency power. Standby power loads include smoke control systems; certain hazardous material operations; smokeproof enclosure systems; illumination; heating, ventilating, and air-conditioning (HVAC) systems; refrigeration; and sewage pumps. Standby power systems must provide power within 60 seconds of failure of primary power.

Sources of power for emergency power systems (NFPA 70, Section 701-11) include storage batteries, generators, uninterruptible power supplies and separate services. Sources of power for standby systems include those allowed for emergency systems plus a source that is taken from a point of connection ahead of the normal service disconnecting means.

NFPA 110 addresses the performance criteria and “nuts and bolts” of emergency and standby power systems and separates them into types, classes and levels relative to maximum response time, minimum required operation time and life safety importance factor, respectively. NFPA 111 addresses stored emergency power supply systems and is similar in coverage to NFPA 110. Stored energy systems (ESS) typically rely on batteries that store chemical energy.

1203.1.4 Load transfer. Emergency power systems shall automatically provide secondary power within 10 seconds after primary power is lost, unless specified otherwise in this code. Standby power systems shall automatically provide secondary power within 60 seconds after primary power is lost, unless specified otherwise in this code.

C These provisions are consistent with NFPA 70 and provide clarity to the fire code official regarding one of the key differences between standby and emergency power. This is discussed in more detail in the commentary to Section 1203.1.3.

1203.1.5 Load duration. Emergency power systems and standby power systems shall be designed to provide the required power for a minimum duration of 2 hours without being refueled or recharged, unless specified otherwise in this code.

C In order to properly design emergency and standby power systems, the minimum load duration must be known. This section provides a default minimum 2-hour duration for systems unless another load duration is specified. For instance, in-building emergency responder communication enhancement systems are required to provide standby power for 12 hours.

1203.1.6 Uninterruptable power source. An uninterrupted source of power shall be provided for equipment where required by the manufacturer's instructions, the listing, this code or applicable referenced standards.

C In some cases, an uninterruptable power supply is required. Uninterruptible power supplies are different from emergency and standby power supplies in that there is no transfer time allowed. This is necessary for equipment with volatile memories or that is sensitive to any loss of power. An uninterruptible power source must be provided if required by the equipment manufacturer's instructions, the listing of the equipment, the code or applicable referenced standards such as NFPA 72.

1203.1.7 Interchangeability. Emergency power systems shall be an acceptable alternative for installations that require standby power systems.

C An emergency power system is more restrictive than a standby power system primarily due to the faster transfer time. This section provides code language that would specifically permit such a substitution. It will result in a faster-reacting system, which will only improve performance.

1203.1.8 Group I-2 occupancies. In Group I-2 occupancies located in flood hazard areas established in Section 1612.3 of the *International Building Code* where new essential electrical systems are installed, and where new essential electrical system generators are installed, the systems and generators shall be located and installed in accordance with ASCE 24. Where connections for hook up of temporary generators are provided, the connections shall be located at or above the elevation required in ASCE 24.

C Essential electrical systems and essential system generators installed in healthcare facilities in areas subject to flooding have operationally failed under flood conditions, even though buildings were designed to meet the structural loads associated with conditions of flooding. Emergency electrical systems and standby power generators are necessary for post-flood life safety and healthcare and for other occupancies and uses. ASCE 24 provides the designer with the minimum requirements and expected performance for the design and construction of buildings and structures in flood hazard areas, including the installation of attendant utilities and equipment and electrical systems. Group I-2 occupancies are assigned Flood Design Class 4 and are required to be elevated or protected to the base flood elevation plus 2 feet or the 500-year flood elevation, whichever is higher. See the commentary to IBC Section 1612 for additional information.

1203.1.9 Maintenance. Existing installations shall be maintained in accordance with the original approval and Section 1203.4.

C Existing generators must be properly maintained to continue to serve the purpose for which they were installed. This section ensures that maintenance. The maintenance level is linked to the original installation approval. Section 1203.4 provides detailed maintenance requirements.

1203.2 Where required. Emergency and standby power systems shall be provided where required by Sections 1203.2.1 through 1203.2.19.

C Fires or natural events can cause the loss of utility power as a result of either damage to equipment and wiring or from firefighters shutting off power to eliminate sources of ignition and danger to personnel.

Sections 1203.2.1 through 1203.2.19 list locations throughout buildings and facilities where emergency or standby power is required. In some cases, the requirements are occupancy- or use-specific; in others they are system- or equipment-specific. Generally, the requirements are related to life safety-oriented systems in buildings, such as a fire alarm system or elevators used for the assisted rescue of those with disabilities. Emergency and standby power systems have different characteristics, and subsequent sections will require one or the other (see commentary, Section 1203.1.3).

In some cases, these subsections define the requirements for standby or emergency power, while in others they serve as pointers to the requirements.

1203.2.1 Ambulatory care facilities. Essential electrical systems for ambulatory care facilities shall be in accordance with Section 422.6 of the *International Building Code*.

C This section is a pointer to IBC Section 422.6 for requirements for essential electrical systems for ambulatory care facilities. Section 422.6 requires compliance with NFPA 99 for essential electrical systems. NFPA 99 provides an assessment tool for

determining what constitutes essential electrical systems. It provides a risk-based approach to determine the need for an essential electrical system, what class system is required and general design requirements for each type of system.

1203.2.2 Elevators and platform lifts. Standby power shall be provided for elevators and platform lifts as required in Sections 604.3, 1009.4.1 and 1009.5.

C Elevators can be a component of an accessible means of egress in accordance with Section 1009.4 and must, therefore, be dependable at all times. Without backup power, an elevator could be a dead end for someone with physical disabilities trying to egress a building (see commentary, Section 1009.4). Note that the intent of such standby power in this instance is focused on assisted rescue by the fire department and not self-evacuation.

Platform lifts are allowed to be used as part of an accessible means of egress only where they are allowed as part of an accessible route, in accordance with IBC Section 1109.8, Items 1 through 10. Where this is the case, the platform lift could be a dead end for someone with physical disabilities if power to the platform lift were to be lost; therefore, standby power is required, in accordance with Section 1203 or ASME A18.1.

The reference to Section 604.3 addresses how standby power is to be provided to elevators where required elsewhere. These requirements are the same as those found in IBC Section 3003. Primarily, these requirements apply to high-rise buildings (see commentary, IBC Section 403.4.8.3). Also note that the IBC has specific standby power requirements in Sections 3007.8 and 3008.8 for fire service access elevators and occupant evacuation elevators, respectively. These requirements are more restrictive than those found in Section 604.3 of the code and Section 3003 of the IBC (see commentary, Section 604.3).

1203.2.3 Emergency responder communication coverage systems. Standby power shall be provided for in-building, two-way emergency responder communication coverage systems as required in Section 510.4.2.3. The standby power supply shall be capable of operating the in-building, two-way emergency responder communication coverage system at 100 percent system operation capacity for a duration of not less than 12 hours.

C Loss of power is likely during an emergency situation and, in many cases, the fire department and emergency responders will require the use of the in-building two-way emergency responder communication enhancement system for an extended amount of time. Power can be supplied by dedicated standby batteries or 2 hours of batteries and connection to the standby power generator.

1203.2.4 Emergency voice/alarm communication systems. Standby power shall be provided for emergency voice/alarm communication systems in accordance with NFPA 72.

C Emergency voice/alarm communication systems are required to include an emergency power source in accordance with NFPA 72. NFPA 72 requires the system to remain powered for 24 hours and be able to operate under alarm for 15 minutes. High-rise buildings, Group A occupancies and covered malls are examples of occupancies and building types that require emergency voice/alarm communication systems.

1203.2.5 Exhaust ventilation. Standby power shall be provided for mechanical exhaust ventilation systems as required in Section 1207.6.1.2.1. The system shall be capable of powering the required load for a duration of not less than 2 hours.

C This section requires at least 2 hours of standby power to operate the exhaust system for energy storage systems that require mechanical ventilation. These particular ESS are those prone to the generation of hydrogen gas. These systems can go into thermal runaway and continue to generate flammable gases. Section 1207.6.1.2.1 is essentially a pointer to this section. Table 1207.6 indicates which types of ESS require ventilation.

1203.2.6 Exit signs. Emergency power shall be provided for exit signs as required in Section 1013.6.3. The system shall be capable of powering the required load for a duration of not less than 90 minutes.

C Emergency power is warranted for exit signage illumination since guiding occupants to the exits is a critical life safety function during any emergency where occupants are required to relocate or exit the building. Note that the power system must be available for only 90 minutes versus 2 hours as required in Section 1203.1.5 (see commentary, Section 1013.6.3).

1203.2.7 Gas detection systems. Emergency power shall be provided for *gas detection systems* where required by Sections 1203.2.10 and 1203.2.17. Standby power shall be provided for *gas detection systems* where required by Sections 916.5 and 1207.6.1.2.4.

C This section notes that such emergency power requirements originate both in the hazardous material provisions in Chapter 50 and associated hazardous materials chapters, such as those addressing toxic and highly toxic materials. Additionally, semiconductor facilities as addressed in Chapter 27 require gas detection for higher hazard materials such as toxic and highly toxic gases and flammable gases. Depending on the system, standby power or emergency power is required to reduce the risk of fire from a gas leak during a time of loss of power to the building. Note that as an alternative to such power supplies, Section 916.5 states that the gas detection system can initiate a trouble signal at an approved location if there is a loss of power. The intent being this will cause the necessary intervention to address the lack of power.

1203.2.8 Group I-2 occupancies. Essential electrical systems for Group I-2 occupancies shall be in accordance with Section 407.11 of the *International Building Code*.

C Emergency power systems are required to comply with NFPA 99 by the Center for Medicare/Medicaid Services (CMS) for a federally licensed health care facility to receive federal reimbursement funds. IBC Section 407.11 provides a direct reference to NFPA 99 for the design and construction of emergency power systems in Group I-2. See the commentary to IBC Section 407.11.

1203.2.9 Group I-3 occupancies. Power-operated sliding doors or power-operated locks for swinging doors in Group I-3 occupancies shall be operable by a manual release mechanism at the door. Emergency power shall be provided for the doors and locks.

Exceptions:

1. Emergency power is not required in facilities where provisions for remote locking and unlocking of occupied rooms in Occupancy Condition 4 are not required as set forth in the *International Building Code*.
2. Emergency power is not required where remote mechanical operating releases are provided.

C In an emergency situation involving power loss, occupants in detention and correctional facilities are at the mercy of door-locking mechanisms and those who control such locks; thus, emergency power is warranted. For a discussion of Exception 1, see the commentaries to IBC Sections 408.4.1 and 408.4.2. Exception 2 recognizes that the use of mechanical operating releases controlled remotely eliminates the need for emergency power.

1203.2.10 Hazardous materials. Emergency and standby power shall be provided in occupancies with hazardous materials as required in the following sections:

1. Sections 5004.7 and 5005.1.5 for hazardous materials.
2. Sections 6004.2.2.8 and 6004.3.4.2 for highly toxic and toxic gases.
3. Section 6204.1.11 for *organic peroxides*.

C Where hazardous materials and processes are housed, occupant safety could be dependent on one or more ventilation, treatment, temperature control, alarm or detection system. Thus, emergency or standby power is required, depending on the load. These requirements are typically associated with situations where the maximum allowable quantities (MAQs) are exceeded. The focus is on where mechanical ventilation, treatment systems, temperature control, alarm, detection or other electrically operated systems are required. Note that fail-safe engineered systems are recognized as an alternative to standby or emergency power. See the commentaries to code sections referenced in Items 1 through 3.

1203.2.11 High-rise buildings. Standby power and emergency power shall be provided for high-rise buildings as required in Section 403 of the *International Building Code*, and shall be in accordance with Section 1203.

C Occupants of high-rise buildings are at a greater risk due to longer egress travel times, more difficulty in firefighter access and the danger of vertical spread of fire and upward smoke migration. In accordance with this chapter and IBC Section 403.4.8, some loads in a high-rise building will require standby power and some will require emergency power (see commentary, IBC Section 403.4.8).

1203.2.12 Special purpose horizontal sliding doors. Standby power shall be provided for horizontal sliding doors as required in Section 1010.3.3. The standby power supply shall have a capacity to operate not fewer than 50 closing cycles of the door.

C Power-operated doors could be an obstruction to egress if the primary power supply fails; therefore, standby power is required to maintain door operation (see commentary, Section 1010.3.3).

1203.2.13 Hydrogen fuel gas rooms. Standby power shall be provided for hydrogen fuel gas rooms as required by Section 5808.7.

C Ventilation and gas detection are critical to the ongoing safe operation of a hydrogen fuel gas room. Therefore, both ventilation and gas detection systems are required by Section 5808.7 to be provided with standby power (see commentary, Section 5808.7).

1203.2.14 Laboratory suites. Standby or emergency power shall be provided in accordance with Section 5004.7 where *laboratory suites* are located above the sixth story above *grade plane* or located in a story below *grade plane*.

C This is one component of the safety features that allows the use of laboratory suites as opposed to traditional control areas. Standby or emergency power is triggered where responding to an emergency becomes more difficult due to height above or below grade. In this case, the trigger is six stories above grade plane or on any story below grade plane. See the commentary to Section 3804.1.1.6.

1203.2.15 Means of egress illumination. Emergency power shall be provided for *means of egress* illumination in accordance with Sections 1008.2.4 and 1104.5.1.

C The path of travel to all exits must be illuminated to guide occupants and allow for safe egress. Without proper illumination, the ability of people to find their way to an exit may be hampered significantly. Therefore, emergency power for illumination is warranted (see commentary, Section 1008.3).

1203.2.16 Membrane structures. Standby power shall be provided for auxiliary inflation systems in permanent membrane structures in accordance with Section 2702 of the *International Building Code*. Auxiliary inflation systems shall be provided in temporary air-supported and air-inflated membrane structures in accordance with Section 3103.9.4.

C Air-supported and air-inflated structures could collapse on occupants if inflation systems fail. Section 3102.8.1.1 of the IBC requires redundant inflation equipment in case the primary equipment fails. In addition, such structures require standby power for 4 hours. Temporary air-supported and air-inflated membrane structures also require auxiliary power. However, the auxiliary power is not specifically standby power. Section 3103.9.4 sets out specific criteria as to how this is to be provided. The time it takes to provide this power is 60 seconds, which is the same transfer time required for standby power in Section 1203.1.4.

1203.2.17 Semiconductor fabrication facilities. Emergency power shall be provided for semiconductor fabrication facilities as required in Section 2703.15.

C Where hazardous materials are utilized in Group H-5 hazardous production materials (HPM) facilities, many systems are depended on to protect the occupants from exposure, including exhaust/ventilation systems, gas cabinet exhaust systems, gas detection systems, alarm systems and suppression systems. Loss of power would endanger the occupants; thus, emergency power is essential for these occupancies. For the complete list of required systems, see Section 2703.15 and its commentary.

1203.2.18 Smoke control systems. Standby power shall be provided for smoke control systems as required in Section 909.11.

C Smoke control systems are intended to maintain a tenable environment in certain buildings to allow the occupants ample time to evacuate or relocate to protected areas. As such, smoke control systems are life safety systems and must be dependable (see commentary, Section 909.11). Such systems are more likely to be needed during a power loss than typical building systems such as HVAC. Note that Section 909.11 simply references Section 1203 for standby power requirements, which would default to 2 hours.

1203.2.19 Underground buildings. Emergency and standby power shall be provided in underground buildings as required in Section 405 of the *International Building Code* and shall be in accordance with Section 1203.

C These structures are analogous to inverted high-rise buildings. Due to their configuration, the concerns are similar to a high-rise but made even more challenging by being underground. The need for certain life safety functions such as smokeproof enclosures and smoke control systems is critical. In addition, many fire safety systems such as fire alarm, emergency voice/alarm communication, fire pumps, egress illumination and elevator car lighting are critical. Even a simple power outage without another emergency makes way finding necessary through proper means of egress illumination and exit signage. IBC Section 405.8.1 requires standby power for specified loads and IBC Section 405.8.2 requires emergency power for specified loads (see commentary, IBC Sections 405.8.1 and 405.8.2).

1203.3 Critical circuits. Required *critical circuits* shall be protected using one of the following methods:

1. Cables used for survivability of required *critical circuits* shall be *listed* in accordance with UL 2196 and shall have a *fire-resistance rating* of not less than 1 hour.
2. Electrical circuit protective systems shall have a *fire-resistance rating* of not less than 1 hour. Electrical circuit protective systems shall be installed in accordance with their listing requirements.
3. Construction having a *fire-resistance rating* of not less than 1 hour.

C Several sections specify the need for similar protection, including fire pumps (Section 913.2.2), fire service access elevators (IBC Section 3007.8.1) and occupant evacuation elevators (IBC Section 3008.8.2). The required rating varies from 1 to 2 hours, based on the application. Smokeproof enclosures offer these concepts as alternatives to requiring such control wiring or powering wiring be placed on the exterior, within the rated smokeproof enclosure or protected by rated building construction. Note that smokeproof enclosures would require a 2-hour fire-resistance rating.

This section presents three options for the protection of critical circuits.

The first option is listing to UL 2196 with a rating of 1 hour. UL 2196 is the ANSI-approved standard for tests of fire-resistive cables. NFPA 20 (fire pumps) and NFPA 72 (fire alarms) include selective survivability requirements to ensure integrity of certain critical circuits.

The second option is an electrical circuit protective system with a fire-resistance rating of 1 hour, installed in accordance with the system's listing. NFPA 70 does not specify the applicable standard in the mandatory provisions of the code but recognizes electrical circuit protective systems as alternatives to listed cables. An electrical circuit protective system is a field assembly of components that must be installed according to the listing requirements and manufacturer's instructions in order to maintain the listing for the system. There are more than two dozen electrical circuit protective systems listed in the *UL Fire Resistance Directory*.

The third option is simply to protect the critical circuits in 1-hour fire-resistance-rated construction. This would serve the same intent of protection as the first two options.

1203.4 Maintenance. Emergency and standby power systems shall be maintained in accordance with NFPA 110 and NFPA 111 such that the system is capable of supplying service within the time specified for the type and duration required.

C This section introduces requirements for maintenance of all elements of emergency and standby power systems. The two standards referenced are for emergency and standby power systems and stored electrical energy emergency and standby power systems, respectively. NFPA 110 is geared toward power sources, such as diesel-driven generators, while NFPA 111 is geared toward power supplies, such as energy storage systems (ESS). See also Section 1203.1.9, which requires the same system maintenance as was originally approved.

1203.4.1 Group I-2 and ambulatory care facilities. In Group I-2 occupancies and ambulatory care facilities, emergency and standby power systems shall be maintained in accordance with NFPA 99.

C There are special requirements in NFPA 99 for the maintenance and inspection of generators installed at hospitals, nursing homes and ambulatory care facilities. In addition to these special requirements, NFPA 99 also references NFPA 110.

1203.4.2 Schedule. Inspection, testing and maintenance of emergency and standby power systems shall be in accordance with an *approved* schedule established upon completion and approval of the system installation.

C Standby power and emergency power are useful only if they continue to work over the life of both the building and its associated equipment; therefore, this section specifically focuses on the maintenance of such systems. The primary specifications of these secondary power supplies are that they be able to supply power within the specified length of time.

This section requires that a specific schedule be created at the completion of the installation of the power system to encourage regular maintenance.

1203.4.3 Records. Records of the inspection, testing and maintenance of emergency and standby power systems shall include the date of service, name of the servicing technician, a summary of conditions noted and a detailed description of any conditions requiring correction and what corrective action was taken. Such records shall be maintained.

C Documentation of maintenance is key in that it highlights what was inspected and tested and where potential problems exist. It also provides a level of accountability by providing information about the inspector and inspection agency for future reference. These documents must be made available to the fire code official on request.

As a result of a review of Recommendation 2(c) of the National Institute of Standards and Technology (NIST) Charleston, South Carolina Sofa Superstore Fire Report, changes were made to Section 110.3 and throughout the code to comprehensively address record-keeping requirements.

1203.4.4 Switch maintenance. Emergency and standby power system transfer switches shall be included in the inspection, testing and maintenance schedule required by Section 1203.4.2. Transfer switches shall be maintained free from accumulated dust and dirt. Inspection shall include examination of the transfer switch contacts for evidence of deterioration. When evidence of contact deterioration is detected, the contacts shall be replaced in accordance with the transfer switch manufacturer's instructions.

C One of the most important features of emergency and standby power systems is the ability for the primary power to be switched to the secondary power supply within the specified time; therefore, this section pays specific attention to the long-term reliability of the transfer switches. This includes inspection for cleanliness and signs of deterioration.

1203.5 Operational inspection and testing. Emergency power systems, including all appurtenant components, shall be inspected and tested under load in accordance with NFPA 110 and NFPA 111.

Exception: Where the emergency power system is used for standby power or peak load shaving, such use shall be recorded and shall be allowed to be substituted for scheduled testing of the generator set, provided that appropriate records are maintained.

C This section requires that emergency power systems be tested and inspected as specified in NFPA 110 and 111. There is an exception to testing emergency power systems when they are used either for peak power periods or for standby power. Use during peak hours means that on a fairly regular basis the power supply will be tested. In terms of using emergency power for standby power, the capacity of both types of secondary power systems is the same. The two differ on when they will activate; therefore, testing the system as a standby power supply is adequate when the emergency power system is used as standby power.

1203.5.1 Group I-2 and ambulatory care facilities. In Group I-2 occupancies and ambulatory care facilities, emergency and standby power systems shall be inspected and tested under load in accordance with NFPA 99.

C There are special requirements in NFPA 99 for the maintenance and inspection of generators installed at hospitals, nursing homes and ambulatory care facilities. In addition to these special requirements, NFPA 99 also references NFPA 110.

1203.5.2 Transfer switch test. The test of the transfer switch shall consist of electrically operating the transfer switch from the normal position to the alternate position and then return to the normal position.

C This section prescribes the specific sequence of events for testing transfer switches. The switch must move from the normal position to the alternate position and back again. Emergency power must be available within 10 seconds of the loss of primary power, whereas standby power is to be available within 60 seconds. The transfer switches, therefore, must work within the specified time.

1203.6 Supervision of maintenance and testing. Routine maintenance, inspection and operational testing shall be overseen by a properly instructed individual.

C This section requires a minimum level of qualifications for the testing of emergency and standby power systems. Only trained personnel should test and maintain of these systems. This is consistent with Section 110.4.

SECTION 1204—PORTABLE GENERATORS

1204.1 Portable generators. The use, operation and maintenance of portable generators shall comply with this section.

C Portable generators are frequently used to provide power at special outdoor events, at construction sites and during power outages, which often occur during and following natural disasters. According to the Consumer Product Safety Commission (CPSC), among the 864 carbon monoxide fatalities associated with engine-driven tools occurring from 2004 through 2014, 702 (83%) resulted from 523 incidents associated with generators (including five fatalities from three incidents involving fixed location, permanently installed stationary generators).

Note that the data are based on losses primarily in homes; however, such generators are used in many other applications. This section provides basic safety requirements for the use of portable generators.

1204.2 Listing. Portable generators manufactured after January 1, 2021, shall be *listed* and *labeled* in accordance with UL 2201.

C This section requires portable generators to be listed and labeled in accordance with UL 2201 carbon monoxide mitigation requirements, which (1) limit the amount of CO produced by a portable generator and (2) require the portable generator to shut off in elevated CO environments. These requirements were developed based on an analysis of the CPSC CO incident data and are complementary, addressing both indoor misuse and outdoor use incidents. In accordance with UL 2201, the CO emission rate shall not exceed 150 g/h, and the generator shall shut off at any time when there is a peak 400 ppm CO concentration or the average CO concentration is greater than 150 ppm during a rolling 600 seconds during the test.

1204.3 Operation and maintenance. Portable generators shall be operated and maintained in accordance with the manufacturer's instructions.

C This section requires that such generators are used and maintained as required by the manufacturer. This will increase the appropriate and safe use of portable generators. This will avoid improper applications such as attaching directly to a building's power systems with a transfer switch.

1204.4 Grounding. Portable generators shall be grounded in accordance with NFPA 70.

C This is basic safe practice to avoid electrocution hazards.

1204.5 Operating locations. Portable generators shall be operated only outdoors a minimum of 5 feet (1524 mm) from any building openings such as windows and doors or air intakes. Portable generators shall not be operated within buildings or enclosed areas. Additional separation shall be provided for tents, membrane structures and outdoor assembly events as specified in Chapter 31.

C This section is intended to minimize the chance of carbon monoxide poisoning and fires by restricting locations where portable generators can be used.

1204.6 Cords and wiring. Extension cords and temporary wiring used to connect portable generators shall be in accordance with Section 603 and shall be provided with GFCI protection.

C This section is intended to avoid the use of undersized cords, which can result in overcurrent, electrocution and potentially a fire. As an added protection, ground-fault circuit interrupters (GFCI) are added to prevent the power produced from the generator from reaching the appliance or equipment when an imbalance in current is detected.

1204.7 Connections to premise wiring. Connections to a premise wiring system shall comply with all of the following:

1. Power shall not be provided in a manner that "back-feeds" receptacles or the premise wiring system.
2. Connection to a premise served by commercial power shall be made through a *listed* transfer switch installed, used and maintained in accordance with NFPA 70.
3. Connections to buildings not served by commercial power shall comply with NFPA 70.

C This section provides requirements on how power from portable generators is to be provided to premise wiring systems. Back-feeding receptacles is an unsafe practice that can bypass premise overcurrent protection and create a dangerous situation when local commercial power comes back online. It can also endanger utility workers while they are repairing power lines when they are not expecting any power.

1204.8 Refueling. Portable generators shall not be refueled while operating.

C This relates to the prevention of fires from the possible ignition of flammable vapors.

1204.9 Storage and repair. Storage and repair of fuel-fired portable generators shall comply with Section 313.

C This is simply a pointer to Section 313, which already establishes how such equipment should be stored. See the commentary to Section 313.

1204.10 Fire extinguisher. A *listed* portable fire extinguisher complying with Section 906 with a minimum rating of 2-A:20-B:C shall be provided not more than 50 feet (15 240 mm) from the portable generator.

C This section requires a portable fire extinguisher be provided where portable generators are used. The sizing and rating comply with NFPA 10 for the hazard classification associated with portable generators fueled with a flammable liquid (gasoline). This is consistent with portable fire extinguisher requirements in the code for the mitigation of potential fire hazards.

SECTION 1205—SOLAR PHOTOVOLTAIC POWER SYSTEMS

1205.1 General. Solar photovoltaic (PV) systems shall be installed in accordance with the *International Building Code* or *International Residential Code*. The electrical portion of solar PV systems shall be installed in accordance with NFPA 70. Rooftop-mounted solar photovoltaic systems shall be installed in accordance with Sections 1205.2 through 1205.4.3. Ground-mounted solar photovoltaic systems shall comply with Section 1205.5.

C Solar photovoltaic (PV) power systems are among the most popular alternative energy sources. A number of US electric utility power suppliers offer incentives for the installation of PV systems on buildings because such systems offer property owners the ability to generate their own electricity and, in many cases, sell excess electricity back to the utility provider. Such an arrangement is a beneficial to the utility provider because it reduces the provider's power generation demand, which in turn can control the rates for commercial and residential customers. According to the US Energy Information Administration, approximately 320,208 photovoltaic modules shipped in 2006. In 2022, that number was over 31 million. As the number of PV power systems increases, economy of scale will continue to reduce the systems' costs, making them more common on commercial and residential buildings (see Commentary Figure 1205.1).

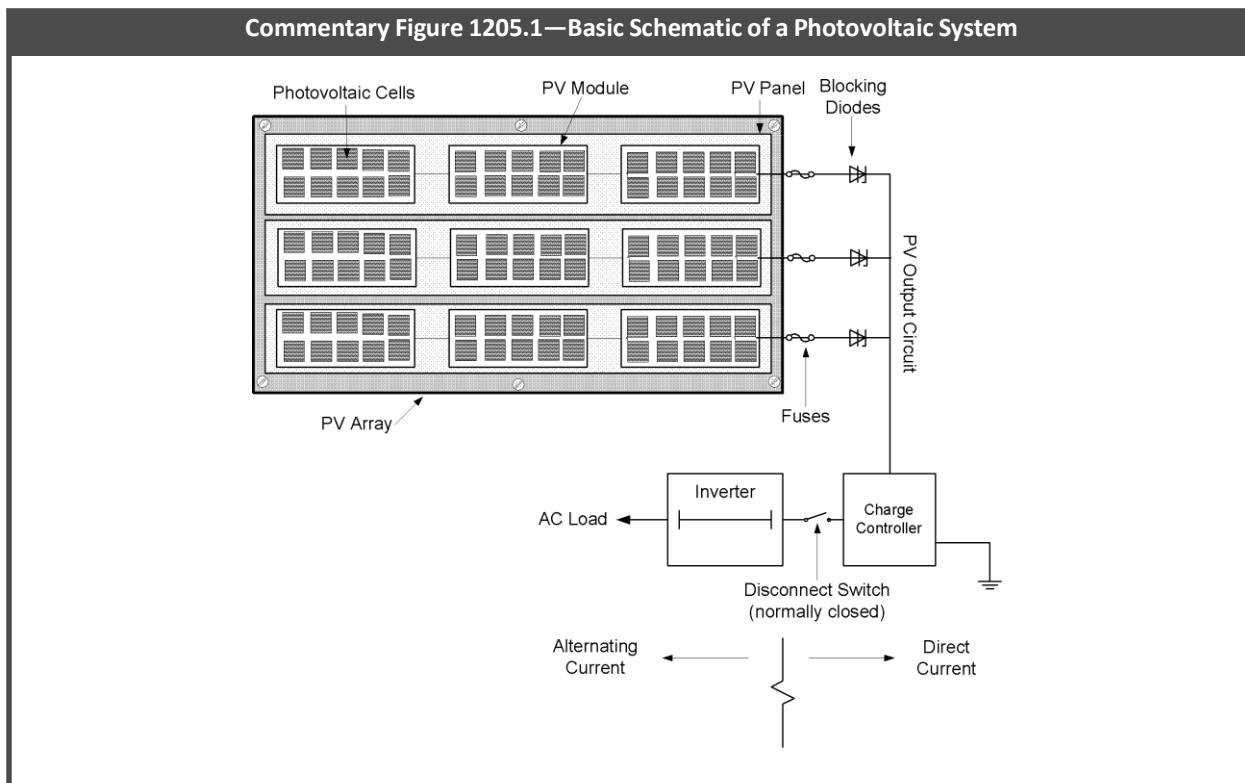
PV systems are designed to convert light energy into direct current (DC) electricity. They have no moving parts and do not contain fluids. The light-to-electricity conversion begins at the PV cell, which is commonly a semiconductor device that generates electricity when exposed to light. To be effective, PV cells are assembled into PV modules, which are then assembled into PV panels. The panels are assembled onto a frame or a flexible substrate, which then can be affixed to the roof of buildings to create a PV array. The PV array and its modules are wired together and generally operate as a series electrical circuit. PV array are required by NFPA 70 to have a fuse or other means of branch-circuit protection to prevent them from being overloaded. While not required, a PV array is commonly equipped with a blocking diode. A blocking diode is analogous to a check valve in a piping system because it limits the direction the electrons can travel. The blocking diode prevents electrical current from one power supply from finding entry into another. In PV systems, the blocking diode protects each individual PV panel if other panels fail and prevents the withdrawal of electricity from the system at night.

The ever-increasing demand for alternative power sources has the potential to pose hazards to emergency responders. The greatest danger for emergency responders operating in proximity to solar energy collection systems is the lack of knowledge needed to operate safely around these systems. Some of the potential hazards associated with PV systems are tripping and falls for firefighters operating on the roof, earlier roof collapse due to the added dead load and electric shock. The provisions of Sections 1205.2 through 1205.5 were developed to provide for the proper installation of PV systems and to address the potential hazards to firefighters. This section requires compliance with the IBC or *International Residential Code*® (IRC®), as applicable, and NFPA 70. Relevant provisions are IBC Sections 1505, 1507 and 3111 and IRC Sections R329, R902, R905 and R907, as applicable. Exterior walls containing BIPV must comply with IBC Section 1411.

The reference to NFPA 70 requires compliance with Section 690 of NFPA 70, which addresses requirements such as marking of equipment and the type of materials that are permitted. Sections 1205.2 and 1205.3 focus primarily on creating pathways where firefighters can perform manual ventilation operations on roofs.

Section 1205.4 addresses solar PV systems that contain the ability to rapidly shut down. There are two types of rapid shutdown addressed by that section. Section 1205.5 addresses ground-mounted solar PV systems.

Commentary Figure 1205.1—Basic Schematic of a Photovoltaic System



1205.2 Access and pathways. Roof access, pathways and spacing requirements shall be provided in accordance with Sections 1205.2.1 through 1205.3.3. Pathways shall be over areas capable of supporting firefighters accessing the roof. Pathways shall be located in areas with minimal obstructions, such as vent pipes, conduit or mechanical equipment.

Exceptions:

1. Detached, nonhabitable Group U structures including, but not limited to, detached garages serving Group R-3 buildings, parking shade structures, carports, solar trellises and similar structures.
2. Roof access, pathways and spacing requirements need not be provided where the *fire code official* has determined that rooftop operations will not be employed.
3. Building-integrated photovoltaic (BIPV) systems where the BIPV systems are *approved*, integrated into the finished roof surface and are *listed* in accordance with UL 3741. The removal or cutting away of portions of the BIPV system during firefighting operations shall not expose a firefighter to electrical shock hazards.

C This section introduces a major requirement of PV system design: to provide access and paths so firefighters can perform the important task of manual ventilation by cutting one or more holes in a building roof. The access also must be safe and available. The pathways need to be able to hold firefighters, useable and free of obstructions. The provisions in Sections 1205.2 through 1205.3.3 address the placement of PV arrays on building roofs. These provisions are broken into two major sections with Sections 1205.2.1 through 1205.2.3 addressing Group R-3 and Sections 1205.3 through 1205.3.3 addressing all other occupancies. It should be noted that these requirements do not apply to buildings regulated by the IRC. See the commentary to Section 102.5 for a discussion of the relationship between this code and the IRC.

Exception 1 recognizes that the installation of PV systems on detached Group U occupancy structures will not require ventilation and therefore does not need to be regulated by this section.

Exception 2 exempts PV arrays from the spacing requirements when the fire chief indicates that vertical manual ventilation practices will not be employed. Conditions that could be considered under this exception include, but are not limited to, proximity and type of adjacent exposures; alternative access opportunities (as from adjoining roofs); adequate ventilation opportunities beneath solar arrays that are sufficiently elevated, spaced or set back from other rooftop equipment; installation of automatic ventilation devices or new technology, methods, or other innovations that ensure adequate fire department access, pathways and ventilation opportunities. Note that a construction permit in accordance with Section 105.6.21 would still be required.

Exception 3 recognizes building-integrated photovoltaic (BIPV) systems, which are integrated into the finished roof surface and do not present significant trip hazards or physical obstacles to equipment, such as ladders, that typically exist with other systems. Portions of these systems, and in some cases the entire roof system, may be listed to UL 3741. BIPV systems available today have been shown through testing to not present electrical hazards to firefighters even when cutting into them during ventilation operations. In order to build confidence in the safety features and performance of

these systems, an evaluation resulting in a listing is required. These systems eliminate the hazard to firefighters and therefore the need to provide for access and pathways for large portions of the roof or in some cases, the entire roof where BIPV systems are installed.

1205.2.1 Solar photovoltaic (PV) systems for Group R-3 buildings. Solar photovoltaic (PV) systems for Group R-3 buildings shall comply with Sections 1205.2.1.1 through 1205.2.3.

Exceptions:

1. These requirements shall not apply to structures designed and constructed in accordance with the *International Residential Code*.
2. These requirements shall not apply to roofs with slopes of 2 units vertical in 12 units horizontal (16.7 percent slope) or less.

C This section introduces the photovoltaic (PV) system access requirements for residential buildings (see Commentary Figure 1205.2.1). Note that the focus is on Group R-3 buildings, which are generally one- and two-family dwellings falling outside the scope of the IRC. As noted in Section 1205.1, the IRC addresses one- and two-family dwellings that fall within the scope of that code. The requirements in IRC Section R329 are very similar to those found in Sections 1205.2.1 through 1205.2.3. Exception 2 exempts low-slope roofs (i.e., a slope that is 2:12 or less) because of the reduced hazards of flatter roofs. Therefore, Sections 1205.2.1.1 through 1205.2.1.3 would not apply to flat roof Group R-3 buildings.

Commentary Figure 1205.2.1—Installation of a PV Array on an R-3 Occupancy



1205.2.1.1 Pathways to ridge. Not fewer than two 36-inch-wide (914 mm) pathways on separate roof planes, from lowest roof edge to ridge, shall be provided on all buildings. Not fewer than one pathway shall be provided on the street or driveway side of the roof. For each roof plane with a photovoltaic array, not fewer than one 36-inch-wide (914 mm) pathway from lowest roof edge to ridge shall be provided on the same roof plane as the photovoltaic array, on an adjacent roof plane or straddling the same and adjacent roof planes.

C Where PV arrays are installed on the roof of a Group R-3 building, a minimum of two 3-foot (914 mm) access pathways must be left from the eave to the roof ridge so firefighters have an unobstructed area in which to cut ventilation openings. In addition, at least one of those paths must be on the street or driveway side for better access to the pathway. Finally, the last sentence is intended to ensure access to the ridge as close as possible if not on the same roof plane of the PV array.

1205.2.1.2 Setbacks at ridge. For photovoltaic arrays occupying 33 percent or less of the plan view total roof area, a setback of not less than 18 inches (457 mm) wide is required on both sides of a horizontal ridge. For photovoltaic arrays occupying more than 33 percent of the plan view total roof area, a setback of not less than 36 inches (457 mm) wide is required on both sides of a horizontal ridge.

C Manual ventilation is most effective when accomplished at the highest portion of the roof above the fire. This section establishes an obstruction-free zone for the roof ridge in order to optimize the area available for effective ventilation, allowing for a 2-foot (609.6 mm) trench cut. A lower setback (18 inches) is allowed for buildings with one-third or less of the roof being covered, based on the fact that other areas may be available for venting. For roofs with more than one-third of the roof covered with PV panels, the setback from the ridge is greater (3 feet). Buildings greater than 33 percent covered likely have fewer options for venting without a greater setback.

1205.2.1.3 Alternative setbacks at ridge. Where an *automatic sprinkler system* is installed within the *dwelling* in accordance with Section 903.3.1.3, setbacks at the ridge shall conform to one of the following:

1. For photovoltaic arrays occupying 66 percent or less of the plan view total roof area, a setback of not less than 18 inches (457 mm) wide is required on both sides of a horizontal ridge.
2. For photovoltaic arrays occupying more than 66 percent of the plan view total roof area, a setback of not less than 36 inches (914 mm) wide is required on both sides of a horizontal ridge.

C This section is essentially the same as Section 1205.2.1.2, with credit provided for an automatic sprinkler system. Instead of a coverage criterion of 33 percent, it is 66 percent. The logic is that a fire is much more likely to be extinguished or controlled by the sprinklers, thus reducing the need to ventilate such structures and thus reducing the need for room to ventilate.

1205.2.2 Emergency escape and rescue openings. Panels and modules installed on Group R-3 buildings shall not be placed on the portion of a roof that is below an emergency escape and rescue opening. A pathway of not less than 36 inches (914 mm) wide shall be provided to the emergency escape and rescue opening.

C This section provides the necessary pathway below emergency escape and rescue openings that are located above a portion of the roof. Without this section it is quite possible PV panels could be installed where access is necessary to emergency escape and rescue openings, thus negating their ability to serve their purpose of escape and rescue.

1205.2.3 Building-integrated photovoltaic (BIPV) systems. Where building-integrated photovoltaic (BIPV) systems are installed in a manner that creates areas with electrical hazards to be hidden from view, markings shall be provided to identify the hazardous areas to avoid for ladder placement. The markings shall be reflective and be visible from grade beneath the eaves or other location approved by the *fire code official*.

Exception: BIPV systems *listed* in accordance with UL 3741, where the removal or cutting away of portions of the BIPV system during firefighting operations have been determined to not expose a firefighter to electrical shock hazards.

C Where BIPV systems are installed on roofs, the areas containing electrical hazards can be hidden from view of firefighters and other responders seeking to access the roof for emergency operations. Some manufacturers produce systems where photovoltaic and nonphotovoltaic tiles have the same finish, preventing any awareness of where the system is located. This section requires markings warning firefighters and others who access the roof as to where the hazards are located and where to avoid ladder placement. The marking must be readily visible day and night and visible from grade to avoid ground ladder placement at an area with an electrical hazard. These markings must be visible in locations where they are most likely to be seen, such as beneath the eaves. This section provides authority to the fire code official to approve alternative locations.

The exception recognizes systems that are specifically designed to address shock hazards and would not require such markings. To comply, the BIPV systems are required to be evaluated and listed in accordance with UL 3741.

1205.3 Other than Group R-3 buildings. Access to systems for buildings, other than those containing Group R-3 occupancies, shall be provided in accordance with Sections 1205.3.1 through 1205.3.3.

Exception: Where it is determined by the *fire code official* that the roof configuration is similar to that of a Group R-3 occupancy, the residential access and ventilation requirements in Sections 1205.2.1.1 through 1205.2.1.3 are a suitable alternative.

C This section introduces PV system access requirements for other than Group R-3 buildings. The exception recognizes that nonresidential buildings are often constructed in a residential configuration and authorizes the fire code official to allow the provisions for Group R-3 construction styles to be used. The access and ventilation requirements are not occupancy specific but are affected most by the construction configuration.

1205.3.1 Perimeter pathways. There shall be a minimum 6-foot-wide (1829 mm) clear perimeter around the edges of the roof.

Exception: Where either axis of the building is 250 feet (76 200 mm) or less, the clear perimeter around the edges of the roof shall be permitted to be reduced to a minimum width of 4 feet (1219 mm).

C PV arrays installed on the roofs of nonresidential buildings must be located so they create a minimum 6-foot (1829 mm) clearance between the PV array and roof edge so firefighters have an unobstructed area in which to move and operate when cutting ventilation openings. The exception provides some flexibility for smaller buildings where less access to the roof is needed; therefore, a reduced width of 4 feet is acceptable.

1205.3.2 Interior pathways. Interior pathways shall be provided between array sections to meet the following requirements:

1. Pathways shall be provided at intervals not greater than 150 feet (45 720 mm) throughout the length and width of the roof.
2. A pathway not less than 4 feet (1219 mm) wide in a straight line to roof standpipes or ventilation hatches.
3. A pathway not less than 4 feet (1219 mm) wide around roof access hatches, with not fewer than one such pathway to a parapet or roof edge.

C Pathways must be established in the design of the PV systems and meet the requirements listed in this section. The first item denotes that a straight pathway from the roof edge must be provided every 150 feet. Therefore, a 300-foot-long build-

ing would require a single pathway in the middle. Note that Section 1205.3.1 requires a 6-foot clear perimeter along the edge of the roof. Straight pathways with a 4-foot (1290 mm) clearance to tripping and other hazards help provide a safe working environment for personnel assigned to rooftop operations. Additionally, access to roof standpipes and ventilations hatches is necessary.

1205.3.3 Smoke ventilation. The solar installation shall be designed to meet the following requirements:

1. Where nongravity-operated smoke and heat vents occur, a pathway not less than 4 feet (1219 mm) wide shall be provided bordering all sides.
2. Where gravity-operated dropout smoke and heat vents occur, a pathway not less than 4 feet (1219 mm) wide on not fewer than one side.
3. Smoke ventilation options between array sections shall be one of the following:
 - 3.1. A pathway not less than 8 feet (2438 mm) wide.
 - 3.2. A pathway not less than 4 feet (1219 mm) wide bordering 4-foot by 8-foot (1219 mm by 2438 mm) venting cutouts every 20 feet (6096 mm) on alternating sides of the pathway.

C This section establishes design options for providing adequate space among and around the PV arrays for safe fire department ventilation operations.

Nongravity-operated smoke and heat vents will require more intervention from the fire department; therefore, more access to those vents will be needed.

Note that gravity-operated smoke and heat vents are only required to provide access on one side. Gravity-operated smoke and heat vents will automatically provide ventilation if needed.

This section also requires smoke ventilation between array sections and provides two options. The first option is simply to provide 8-foot-wide pathways between arrays, and the second option allows a 4-foot-wide pathway where 4-foot by 8-foot venting cutouts are provided every 20 feet.

1205.4 Buildings with rapid shutdown. Buildings with rapid shutdown solar photovoltaic systems shall have permanent labels in accordance with Sections 1205.4.1 through 1205.4.3.

C Rapid shutdown features of solar PV systems are beneficial to firefighters in that they can more quickly and easily reduce hazards and fight the fire. Shutting down the system more quickly reduces the live electrical aspects of the system that could injure or kill firefighters. However, it needs to be clear what is being shut down on the system. It must be understood whether it is both the array and conductors or only the conductors. Additionally, signage in general is necessary to determine if there is more than one shut down for the system and if portions of the system are not rapid shutdown. This section is not intended to require rapid shutdown but instead to appropriately provide signage to understand where rapid shutdown capability exists. These requirements are consistent with NFPA 70.

1205.4.1 Rapid shutdown type. The type of solar photovoltaic system rapid shutdown shall be labeled with one of the following:

1. For solar photovoltaic systems that shut down the array and the conductors leaving the array, a label shall be provided. The first two lines of the label shall be uppercase characters with a minimum height of $\frac{3}{8}$ inch (10 mm) in black on a yellow background. The remaining characters shall be uppercase with a minimum height of $\frac{3}{16}$ inch (5 mm) in black on a white background. The label shall be in accordance with Figure 1205.4.1(1) and state the following:

SOLAR PV SYSTEM EQUIPPED WITH
RAPID SHUTDOWN. TURN RAPID
SHUTDOWN SWITCH TO THE "OFF"
POSITION TO SHUT DOWN PV SYSTEM
AND REDUCE SHOCK HAZARD IN ARRAY.

2. For photovoltaic systems that only shut down conductors leaving the array, a label shall be provided. The first two lines of the label shall be uppercase characters with a minimum height of $\frac{3}{8}$ inch (10 mm) in white on a red background and the remaining characters shall be capitalized with a minimum height of $\frac{3}{16}$ inch (5 mm) in black on a white background. The label shall be in accordance with Figure 1205.4.1(2) and state the following:

THIS SOLAR PV SYSTEM EQUIPPED
WITH RAPID SHUTDOWN. TURN RAPID
SHUTDOWN SWITCH TO THE "OFF"
POSITION TO SHUT DOWN CONDUCTORS
OUTSIDE THE ARRAY. CONDUCTORS
WITHIN ARRAY REMAIN ENERGIZED IN SUNLIGHT.

C Rapid shutdown can apply to both the conductors and the array or simply the conductors. Distinguishing this difference is important, as one PV system remains electrically live while in the other type of system all electrical is shut down. There are two types of signs that can be used, depending on the system [see Figures 1205.4.1(1) and 1205.4.1(2)].

FIGURE 1205.4.1(1)—LABEL FOR SOLAR PV SYSTEMS THAT REDUCE SHOCK HAZARD WITHIN ARRAY AND SHUT DOWN CONDUCTORS LEAVING ARRAY

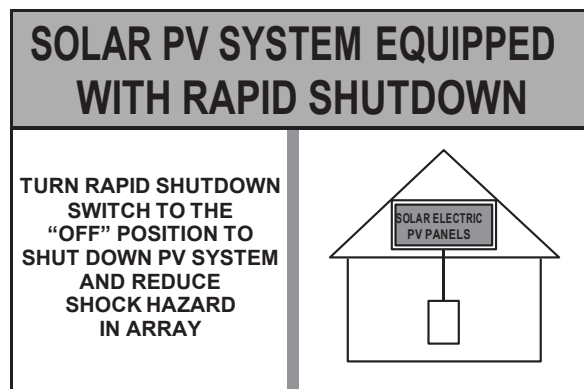
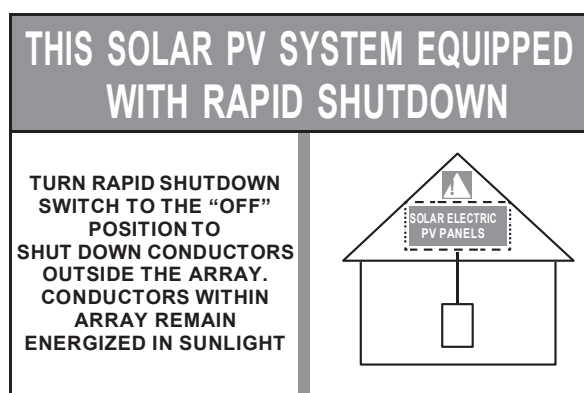


FIGURE 1205.4.1(2)—LABEL FOR SOLAR PV SYSTEMS THAT ONLY SHUT DOWN CONDUCTORS LEAVING THE ARRAY



1205.4.1.1 Diagram. The labels in Section 1205.4.1 shall include a simple diagram of a building with a roof. Diagram sections in red signify sections of the solar photovoltaic system that are not shut down when the rapid shutdown switch is turned off.

C In addition to the signage, a diagram must also be provided. This helps to better identify through color which portions of the systems will remain live after rapid shutdown.

1205.4.1.2 Location. The rapid shutdown label in Section 1205.4.1 shall be located not greater than 3 feet (914 mm) from the service disconnecting means to which the photovoltaic systems are connected, and shall indicate the location of all identified rapid shutdown switches if not at the same location.

C Signage and diagrams are only helpful if located close to the disconnecting means. This section specifies a maximum distance of 3 feet that these signs can be located in association with the service disconnect. Note that buildings with multiple rapid shutdown switches need to identify where all the switches are located.

1205.4.2 Buildings with more than one rapid shutdown type. Solar photovoltaic systems that contain rapid shutdown in accordance with both Items 1 and 2 of Section 1205.4.1 or solar photovoltaic systems where only portions of the systems on the building contain rapid shutdown, shall provide a detailed plan view diagram of the roof showing each different photovoltaic system and a dotted line around areas that remain energized after the rapid shutdown switch is operated.

C Buildings can have two types of rapid shutdown where either the array and conductors are shut down or only the conductors. Also note it is possible to have a building with photovoltaics with some form of rapid shutdown and portions without. Due to the potential hazards to firefighters, these differences must be provided in a detailed plan view that shows which areas are still energized.

1205.4.3 Rapid shutdown switch. A rapid shutdown switch shall have a label located not greater than 3 feet (914 mm) from the switch that states the following:

RAPID SHUTDOWN SWITCH
FOR SOLAR PV SYSTEM

C This section requires that the actual switch be labeled and such label cannot be further than 3 feet from the rapid shutdown switch. The requirements of this section are intended to assist in providing information to firefighters to maintain safety while fighting fires that contain photovoltaics.

1205.5 Ground-mounted photovoltaic panel systems. Ground-mounted photovoltaic panel systems shall be installed in accordance with this section. Setback requirements shall not apply to ground-mounted, free-standing photovoltaic arrays.

C This section applies all requirements for rooftop PV systems to ground-mounted PV systems except for the access, pathway and smoke ventilation clearance requirements. Since PV systems are no longer on a roof that potentially needs to be vented, the physical hazards of ground-mounted systems are reduced.

1205.5.1 Vegetation control. A clear, brush-free area of 10 feet (3048 mm) shall be required around the perimeter of the ground-mounted photovoltaic arrays. A maintained vegetative surface or a noncombustible base, *approved* by the *fire code official*, shall be installed and maintained under the photovoltaic arrays and associated electrical equipment installations.

C For visibility and emergency access, a clear zone is required around the arrays. This clearance also reduces the risk of a fire occurring adjacent to the arrays. Some level of vegetation is acceptable as long as it is properly maintained.

SECTION 1206—STATIONARY FUEL CELL POWER SYSTEMS

1206.1 General. *Stationary fuel cell power systems* in new and existing occupancies shall comply with this section.

Exception: The temporary use of a fuel cell-powered electric vehicle to power a Group R-3 or R-4 building while parked shall comply with Section 1206.13.

C This section deals with the regulation of stationary fuel cell power systems based on the fact that they are located in buildings, and the potential hazards they pose need to be addressed. Fuel cell power systems are being used in ever-increasing numbers to meet facility energy needs. Stationary fuel cell power systems generate power through an electrochemical process that combines hydrogen and oxygen to produce electricity. The hydrogen comes from a direct hydrogen source or from any hydrocarbon fuel such as natural gas, gasoline, diesel, or methanol if the fuel cell power system includes integral reforming. The oxygen comes from air around the fuel cell. There are several types of fuel cell systems that require different approval processes. These will be discussed further in Section 1206.3.

The exception points out the allowance in Section 1206.13 to temporarily provide power to a Group R-3 or R-4 occupancy. This allowance is also provided in the IRC for similar structures. See the commentary to Section 1206.13.

1206.2 Permits. Permits shall be obtained for *stationary fuel cell power systems* as set forth in Section 105.6.10.

C Section 105.6.10 requires a construction permit to allow the fire department to become familiar with and sign off on the safety of such systems.

1206.3 Equipment. *Stationary fuel cell power systems* shall comply with the following:

1. *Prepackaged fuel cell power systems* shall be *listed* and *labeled* in accordance with CSA FC 1.
2. The modules and components in a *preengineered fuel cell power system* shall be *listed* and *labeled* in accordance with CSA FC 1 and interconnected to complete the assembly of the system at the job site in accordance with the manufacturer's instructions and the module and component listings.
3. *Field-fabricated fuel cell power systems* shall be *approved* based on a review of the technical report provided in accordance with Section 104.2.2. The report shall be prepared by and bear the stamp of a *registered design professional* and shall include:
 - 3.1. A fire risk evaluation.
 - 3.2. An evaluation demonstrating that modules and components in the fuel cell power system comply with applicable requirements in CSA FC 1.
 - 3.3. Documentation of the fuel cell power system's compliance with applicable NFPA 2 and NFPA 853 construction requirements.

C Three types of fuel cell power systems are defined in Chapter 2. They include:

- Prepackaged fuel cell power systems.
- Preengineering fuel cell power systems.
- Field-fabricated fuel cell power systems.

Each of these systems has a different approach to approval. Prepackaged systems come as a complete unit and simply need to be listed and labeled in accordance with CSA FC1. These systems are ready for installation right from the factory, whereas preengineered and field-fabricated systems require more approvals during installation. The requirements for listing and labeling to CSA FC1 remove the burden to the fire code official of determining compliance with NFPA 853, which is a standard for the design, construction, and installation of stationary fuel cell power systems very similar in content to CSA FC1. Preengineered systems fall between prepackaged and field fabricated. The modules and components are listed and labeled, and the system is approved but must be assembled at the job site. Again, the listing essentially demonstrates compliance with NFPA 853. In addition to listing and labeling preengineered fuel cell power systems, the manufacturer's

instructions must be followed. Field fabricated is a system completely assembled at the job site and the modules are not listed or labeled. Instead, they need to meet several criteria, including demonstrating compliance with CSA FC1. This option allows for newer technologies to be used that have not been designed yet as prepackaged or preengineered. In addition to compliance with CSA FC 1, a risk analysis is necessary. Additionally, hydrogen, a common fuel used for fuel cell power systems, must meet the requirements in NFPA 2 and NFPA 853.

The standard CSA FC 1 is used to investigate and list the stationary fuel cells covered by this section. The construction and performance requirements in that standard address a variety of hazards, including mechanical, electrical, thermal, malfunction, erroneous human intervention and environmental. NFPA 853 includes requirements for the design, construction and installation of stationary fuel cell power systems.

1206.4 Installation. *Stationary fuel cell power systems* shall be installed and maintained in accordance with NFPA 70 and NFPA 853, the manufacturer's installation instructions, and the listing. *Stationary fuel cell power systems* fueled by hydrogen shall be installed and maintained in accordance with NFPA 2 and NFPA 70, the manufacturer's installation instructions and the listing.

C This section is provided to ensure that such systems are properly installed and maintained, regardless of type. Several standards are called up for this process.

1206.5 Residential use. *Stationary fuel cell power systems* shall not be installed in Group R-3 and R-4 buildings, or *dwelling units* associated with Group R-2 buildings unless they are specifically *listed* for residential use.

Exception: The temporary use of a fuel cell-powered electric vehicle to power a Group R-3 or R-4 building while parked shall comply with Section 1206.13.

C Systems are listed for both residential and nonresidential use based on the requirements in NFPA 853 and CSA FC1. This section is provided to ensure the appropriate stationary fuel cell power system is installed.

The exception acknowledges the allowance in Section 1206.13 that would allow the use of a vehicle for temporary power for Group R-3 and R-4 occupancies. See the commentary to Section 1206.13.

1206.6 Indoor installations. *Stationary fuel cell power systems* installed in indoor locations shall comply with Sections 1206.6 through 1206.6.2. For purposes of this section, an indoor location includes a roof and 50 percent or greater enclosing walls.

C Systems are listed for either indoor or outdoor use. The 50 percent wall limitation is from NFPA 853, Section 3.3.15.2. This criterion is necessary to differentiate which installations are considered indoor versus outdoor, as the requirements vary.

1206.6.1 Listed. *Stationary fuel cell power systems* installed indoors shall be specifically *listed* and *labeled* for indoor use.

C The purpose of this section is to ensure that indoor systems are listed and labeled appropriately for indoor use. This makes it clear to the fire code official that what is being installed is appropriate for the location and use without requiring verification through additional documentation.

1206.6.2 Separation. Rooms containing *stationary fuel cell power systems* shall be separated from the following occupancies by *fire barriers* or *horizontal assemblies*, or both, constructed in accordance with the *International Building Code*.

1. Group B, F, M, S and U occupancies by 1-hour *fire-resistance-rated* construction.
2. Group A, E, I and R occupancies by 2-hour *fire-resistance-rated* construction.

Exception: *Stationary fuel cell power systems* with an aggregate rating less than 50 kW shall not be required to be separated from other occupancies provided that the systems comply with Section 9.3 of NFPA 853.

C These systems are essentially like an incidental use to the building and separations are provided for consistency with that concept. The separations are similar to those required for ESS in Section 1207.7.4. Note that Group B, F, M, S and U occupancies would require only a 1-hour separation, whereas a 2-hour separation would be required for ESS. Systems less than 50 kW do not require additional separation, as the risk they pose is low.

1206.6.3 Gas detection systems. *Stationary fuel cell power systems* shall be provided with a *gas detection system*. Detection shall be provided in *approved* locations in the fuel cell power system enclosure, the exhaust system or the room that encloses the fuel cell power system. The system shall be designed to activate at a *flammable gas* concentration of not more than 25 percent of the lower flammable limit (LFL).

C Gas detection system requirements include detection locations from UL 853 and activation criteria that are consistent with the requirements of this code. Gas detection is critical with fuels such as hydrogen and is intended for fuel cell power systems using such fuels. The necessary locations for the detection devices will vary based on each installation and system; therefore, the term "approved" is used for their location. Note that NFPA 853 states that the detectors be located based on "leakage sources and fuel type." The intent is not to provide gas detection when liquid fuels are being used. NFPA 853 does require leak detection for liquid fuels.

1206.6.3.1 System activation. The activation of the *gas detection system* shall automatically:

1. Close valves between the gas supply and the fuel cell power system.

2. Shut down the fuel cell power system.
3. Initiate local audible and visible alarms in *approved* locations.

C The initiation of the gas detection creates three specific actions that shut off the gas supply, shut down the fuel cell and provide at least a local alarm to alert someone of the hazard. Note that such actions occur once the gas has reached 25 percent of the lower flammable limit (LFL). Again, gas detection is intended for liquefied or gaseous fuels.

1206.7 Vehicle impact protection. Where *stationary fuel cell power systems* are subject to impact by a motor vehicle, vehicle impact protection shall be provided in accordance with Section 312.

C The provisions in this section protect the system from physical damage to avoid a malfunction of the equipment and the possible resulting hazards. A good example may be a fuel cell in a warehouse space where forklifts are being used (see commentary, Section 312).

1206.8 Outdoor installation. *Stationary fuel cell power systems* located outdoors shall be separated by not less than 5 feet (1524 mm) from the following:

1. Lot lines.
2. Public ways.
3. Buildings.
4. Stored combustible materials.
5. Hazardous materials.
6. High-piled stock.
7. Any portion of a designated *means of egress* system.
8. Other exposure hazards.

C The exposures are different for installations on the exterior, as there are other exposures such as adjacent buildings. The separations are based on, but not identical to, requirements in NFPA 853, Section 5.1.1.

1206.9 Fuel supply. The design, location and installation of the fuel supply for *stationary fuel cell power systems* shall comply with Chapter 53, Chapter 58 and the *International Fuel Gas Code*, based on the particular fuel being supplied to the system.

C This section does not address the fuel supply but instead provides references to the appropriate requirements. The primary fuels are hydrogen and CNG.

1206.10 Manual shutoff. Access to a manual shutoff valve shall be provided for the fuel piping within 6 feet (1829 mm) of any fuel storage tank serving the fuel cell and within 6 feet (1829 mm) of the power system. If the fuel tank and the *stationary fuel cell power system* are less than 12 feet (3658 mm) apart, a single shutoff valve shall be permitted. If the *stationary fuel cell power system* is located indoors, the shutoff valve shall be located outside of the room in which the system is installed, unless otherwise *approved* by the *fire code official*.

C Manual shutoff requirements are based on, but not identical to, requirements in NFPA 853, Section 6.4. Note that Section 6.4 deals with hydrogen specifically, but this section would deal with any fuel being used for the fuel cell. Fuel cells generate electricity but need a fuel source such as hydrogen or CNG. Therefore, shutoff valves must be provided in case of malfunction of the fuel supply at the fuel cell to avoid further hazards.

This section essentially is intended to provide two shutoff valves that are located within 6 feet of the fuel cell and within 6 feet of the fuel source. Some flexibility is given to systems where the fuel source and cell are less than 12 feet apart and thus can share a single shutoff valve. Indoor fuel cells must provide shutoff valves outside the room in which the fuel cell is located. This reduces the risk of entering a space with a flammable gas.

1206.11 Ventilation and exhaust. Ventilation and exhaust for stationary fuel cell power systems shall be provided in accordance with NFPA 853.

C Ventilation, exhaust, fire suppression and gas detection are important parts of a safe installation, and NFPA 853 is referenced for convenience. Chapter 7 of NFPA 853 addresses ventilation and exhaust. Section 7.1.1 of NFPA 853 exempts fuel cell power systems installed outdoors and listed prepackaged or preengineered and matched modular fuel cell power systems that are sealed, provide direct ventilation and exhaust in accordance with the listing and manufacturer. Outdoor installations have more air circulation to reduce the hazard of flammable gas buildup, and preengineered and prepacked installations design ventilation and exhaust into the system. The provisions focus on having separate ventilation and exhaust for the power systems and to interlock the systems in case of failure of the exhaust system.

1206.12 Fire protection . Fire protection systems for stationary fuel cell power system installations shall be provided in accordance with NFPA 853.

C For outdoor installations, NFPA 853 essentially requires appropriate fire hydrants be available to fight a fire. Fuel cell power systems that store flammable and combustible liquids for a fuel source require a minimum of 250 gpm for 2 hours. Where flammable and combustible liquids are not used, the existing yard or city hydrant system would be sufficient. In the case of

this code, compliance with Section 507 is sufficient. Where hydrants are not available, NFPA 853 offers the option of undertaking a risk evaluation to address the protection needs.

Indoor installations in accordance with NFPA 853 require automatic fire protection systems within the liquid fuel pump rooms. Such protection systems need to be interlocked with the fuel cell power system to shut down the system upon activation.

1206.13 Group R-3 and R-4 fuel cell vehicle energy storage system use. The temporary use of the *dwelling unit owner* or occupant's fuel cell-powered electric vehicle to power a Group R-3 or R-4 dwelling while parked in an attached or detached garage or outside shall comply with the vehicle manufacturer's instructions and NFPA 70.

C This section permits the use of fuel cell-powered vehicles as a temporary power supply for smaller residential buildings, essentially one- and two-family homes and townhouses. This same allowance is also found in the IRC. The vehicles must belong to an owner or occupant of the dwelling unit, not a third party. (Third-party applications of mobile ESS are separately regulated by Section 1207.) Additionally, the use of fuel cell-powered vehicles must comply with the vehicle manufacturer's instructions and NFPA 70 to provide for an appropriate level of safety. The vehicle manufacturer's information covers the approved electrical connection on the vehicle itself in accordance with federal standards, and NFPA 70 covers the necessary requirements for the electrical connections to the dwelling. Vehicles capable of being used in this manner are more than a convenience issue; in times of natural disasters with associated power outages, the ability to utilize the energy provided by the owner/occupant's personal vehicle could be critical.

SECTION 1207—ELECTRICAL ENERGY STORAGE SYSTEMS (ESS)

1207.1 General. The provisions in this section are applicable to stationary and mobile electrical energy storage systems (ESS).

Exception: ESS in Group R-3 and R-4 occupancies shall only be required to comply with Section 1207.11 except where Section 1207.11.4 requires compliance with Sections 1207.1 through 1207.9.

C Advancements in battery technologies have introduced a new generation of battery technologies, such as lithium-ion and flow batteries, each with advantages and potential hazards. Examples such as the events in Surprise, Arizona, in 2019 where four firefighters were seriously injured due to an explosion involving an ESS installation are a reminder of why ESS need to be regulated. That particular installation did not have the benefit of more current code requirements (<https://ulfirefighter-safety.org/posts/four-firefighters-injured-in-lithium-ion-battery-energy-storage-system-explosion.html>).

"Energy storage systems (ESS)" are defined in Chapter 2, and the scope of these provisions as noted cover both stationary installations and mobile ESS. These requirements are closely aligned with NFPA 855.

The exception clarifies that ESS in Group R-3 and R-4 occupancies are required to comply with only Section 1207.11, unless the systems become too large and exceed the maximum energy ratings set forth in Section 1207.11.4.

1207.1.1 Utilities and industrial applications. This section shall not apply to capacitors and capacitor equipment for electric utilities and industrial facilities used in applications such as flexible AC transmission (FACTS) devices, filter capacitor banks, power factor correction, and standalone capacitor banks for voltage correction and stabilization. (Material based on NFPA 855 2023 Ed.)

C This section provides certain exceptions for utility and industrial installations. Flexible AC transmission system (FACTS) devices regulate energy transmission by controlling line resistance, electrical phases and voltage. A capacitor bank is a system consisting of several capacitors connected in series or parallel to form an energy storage system. This system helps correct the power factor lag and phase shift in an alternating current power supply, increasing electrical energy transfer efficiency.

These facilities are often regulated by state or federal agencies and overall have a favorable fire safety record. Section 1207 includes exemptions for certain types of ESS under the exclusive control of telecommunications and utility industries for specific applications that have a proven safety record.

1207.1.2 Mobile ESS. Mobile ESS deployed at an electric utility substation or generation facility for 90 days or less shall not add to the threshold values in Table 1207.1.3 for the stationary ESS installation if both of the following conditions apply:

1. The mobile ESS complies with Section 1207.10.
2. The mobile ESS is being used only during periods in which the facility's stationary ESS is being tested, repaired, retrofitted or replaced.

(Material based on NFPA 855 2023 Ed.)

C This section gives flexibility for the use mobile ESS as short-term power sources without affecting the threshold quantities that trigger compliance with Section 1207. This depends on compliance with a set of requirements specific to mobile ESS, and this temporary use must be associated support needed during testing, repairs, retrofits or replacement of other systems at the substation or generation facility.

1207.1.3 Scope. ESS having capacities exceeding the values shown in Table 1207.1.3 shall comply with this section. (Material based on NFPA 855 2023 Ed.)

C This section establishes the applicability of Section 1207 for certain ESS. More specifically this Section calls out Table 1207.1.3, which provides the energy capacity threshold based on battery technology type. More hazardous technologies such as lithium-ion batteries have much lower thresholds than those with a longer safety track record, such as lead-acid and nickel-cadmium technologies.

TABLE 1207.1.3—ENERGY STORAGE SYSTEM (ESS) THRESHOLD QUANTITIES

TECHNOLOGY	ENERGY CAPACITY ^a
Capacitor ESS	3 kWh
Flow batteries ^b	20 kWh
Lead-acid batteries, all types	70 kWh ^c
Lithium-ion batteries	20 kWh
Nickel-cadmium (Ni-Cd), nickel metal hydride (Ni-MH) and nickel zinc (Ni-Zn) batteries	70 kWh
Nonelectrochemical ESS ^d	70 kWh
Other battery technologies	10 kWh
Other electrochemical ESS technologies	3 kWh
Sodium nickel chloride batteries	70 kWh
Zinc manganese dioxide batteries (Zn-MnO ₂)	70 kWh

For SI: 1 kilowatt hour = 3.6 megajoules.

a. Energy capacity is the total energy capable of being stored (nameplate rating), not the usable energy rating. For units rated in amp-hours, kWh shall equal rated voltage times amp-hour rating divided by 1,000.

b. Shall include vanadium, zinc-bromine, polysulfide-bromide and other flowing electrolyte-type technologies.

c. Fifty gallons of lead-acid battery electrolyte shall be considered equivalent to 70 kWh.

d. Covers nonelectrochemical technologies such as flywheel and thermal ESS.

C Table 1207.1.3 lists each type of technology and associated energy capacity. Where those capacities are exceeded, the provisions of Section 1207 apply. This will result in the application of a number of compliance requirements, including commissioning, listings, spacing, fire-resistance-rated separations and limits on maximum allowable quantities.

1207.1.4 Permits. Permits shall be obtained for ESS as follows:

1. Construction permits shall be obtained for stationary ESS installations and for mobile ESS charging and storage installations covered by Section 1207.10.1. Permits shall be obtained in accordance with Section 105.6.6.
2. Operational permits shall be obtained for stationary ESS installations and for mobile ESS deployment operations covered by Section 1207.10.3. Permits shall be obtained in accordance with Section 105.5.14.

C Construction permits are essential as they facilitate the necessary review process for such ESS by the fire code official. These systems can be used in many different applications, each with their own specific hazards to be addressed based on those applications. This section also requires operational permits for both stationary and mobile ESS so that the fire department is aware of potential ongoing hazards in the community.

1207.1.4.1 Communication utilities. Operational permits shall not be required for lead-acid and nickel-cadmium battery systems at facilities under the exclusive control of communications utilities that comply with NFPA 76 and operate at less than 50 voltage alternating current (VAC) and 60 voltage direct current (VDC).

C This section allows operational permits to be omitted for a proven technology already appropriately regulated by communication utilities.

1207.1.5 Construction documents. The following information shall be provided with the permit application:

1. Location and layout diagram of the room or area in which the ESS is to be installed.
2. Details on the hourly *fire-resistance ratings* of assemblies enclosing the ESS.
3. The quantities and types of ESS to be installed.
4. Manufacturer's specifications, ratings and listings of each ESS.
5. Description of energy (battery) management systems and their operation.
6. Location and content of required signage.
7. Details on fire suppression, smoke or fire detection, thermal management, ventilation, exhaust and *deflagration* venting systems, if provided.

8. Support arrangement associated with the installation, including any required seismic restraint.
9. A commissioning plan complying with Section 1207.2.1.
10. A decommissioning plan complying with Section 1207.2.3.
11. A fire safety and evacuation plan in accordance with Section 404.

C Extensive documentation is required to obtain a permit. This helps the fire code official understand the extent and location of the hazard and to better be able to determine compliance with Section 1207. This information is also necessary for planning purposes. These documents specifically require both a commissioning and decommissioning plan, which is critical to overall safety. ESS must be properly installed and tested but also cannot simply be abandoned because of the energy and hazards that may remain.

1207.1.5.1 Utilities applicability. Plans and specifications associated with ESS owned and operated by electric utilities as a component of the electric grid that are considered critical infrastructure documents in accordance with the provisions of the North American Electric Reliability Corporation and other applicable governmental laws and regulations shall be made available to the *fire code official* for viewing based on the requirements of the applicable governmental laws and regulations. (Material based on NFPA 855 2023 Ed.)

C This section authorizes the fire code official to have access to ESS plans and specifications where systems are included as part of the North American Electric Reliability Corporation electrical grid. The intent is to provide the fire code official the ability to plan review energy storage applications and awareness of the various energy storage methods so local emergency responders can prepare suitable pre- and post-incident plans. The key issue is that plans and specifications covered by North American Electric Reliability Corporation rules are not considered publicly accessible documents and the utility is required to secure access to them.

1207.1.6 Hazard mitigation analysis. A failure modes and effects analysis (FMEA) or other *approved* hazard mitigation analysis shall be provided in accordance with Section 104.2.2 under any of the following conditions:

1. Where ESS technologies not specifically identified in Table 1207.1.3 are provided.
2. More than one ESS technology is provided in a single fire area where there is a potential for adverse interaction between technologies.
3. Where allowed as a basis for increasing maximum allowable quantities. See Section 1207.5.2.
4. Where required by the *fire code official* to address a potential hazard with an ESS installation that is not addressed by existing requirements.

C This section requires a hazard mitigation analysis in the form of a failure modes and effects analysis (FMEA) for certain situations. This specifically references Section 104.2.2 for proper analysis and documentation requirements. Item 1 addresses new technologies. This allows for flexibility for innovative technology with appropriate analysis and documentation.

ESS applications are being used in very different ways than in the past, and in some cases, multiple types of systems may be located in a single area within a building. Item 2 requires that where there is more than one type of technology in a single fire area a hazard analysis be conducted to better understand possible adverse conditions. Item 3 allows the maximum allowable quantities in a single fire area to be increased under the condition of conducting a hazard analysis.

Item 4 authorizes the fire code official to require a hazard mitigation analysis for any installation that is not addressed by existing requirements. This addition correlates with Section 1107 requirements for existing facilities.

1207.1.6.1 Fault condition. The hazard mitigation analysis shall evaluate the consequences of the following failure modes. Only single failure modes shall be considered.

1. A thermal runaway condition in a single electrochemical ESS unit.
2. A mechanical failure of a nonelectrochemical ESS unit.
3. Failure of any battery (energy) management system or fire protection system within the ESS equipment that is not covered by the product listing failure mode effects analysis (FMEA).
4. Failure of any required protection system external to the ESS, including but not limited to ventilation (HVAC), exhaust ventilation, smoke detection, fire detection, gas detection or fire suppression system. (Material based on NFPA 855 2023 Ed.)

C This section sets the criteria as to what failure modes need to be evaluated for the hazard analysis. The section notes that only a single failure mode be considered at a time when conducting the hazard analysis. This is similar to the manner in which building and fire codes assume a single ignition point.

The first failure that must be analyzed is a single electrochemical ESS Unit that goes into thermal runaway. The second failure that must be analyzed is failure of a mechanical component of the nonelectrochemical ESS unit.

The third failure mode that is required to be evaluated is the failure of any battery (energy) management system or fire protection system within the ESS equipment that is not already addressed by the product listing failure mode effects analysis (FMEA). The fourth failure mode requires an analysis of the failure of any required protection system external to the ESS, such as a sprinkler, fire detection or ventilation system.

1207.1.6.2 Analysis approval. The *fire code official* is authorized to approve the hazardous mitigation analysis provided that the consequences of the hazard mitigation analysis demonstrate:

1. Fires will be contained within unoccupied ESS rooms or areas for the minimum duration of the fire-resistance-rated separations identified in Section 1207.7.4.
2. Fires involving the ESS will allow occupants or the general public to evacuate to a safe location. (Material based on NFPA 855 2023 Ed.)

C A hazardous mitigation analysis is required under certain circumstances to address uncertainties associated with new technologies and configurations. This section describes the acceptance criteria for the hazard mitigation analysis. For the approval of the hazard analysis, Item 1 requires that it be established that the fault conditions detailed in Section 1207.1.6.1 do not result in the fire spreading out of the room or area of origin based on the required 2-hour duration in Section 1207.7.4. Item 2 includes a performance-based approach that aligns with NFPA 855 and requires that occupants can evacuate to a safe location.

1207.1.6.3 Additional protection measures. Construction, equipment and systems that are required for the ESS to comply with the hazardous mitigation analysis, including but not limited to those specifically described in Section 1207, shall be installed, maintained and tested in accordance with nationally recognized standards and specified design parameters.

C Essentially, this section ensures that the ESS are designed and installed in accordance with industry practice and work with the design parameters approved for the unique installation.

1207.1.7 Large-scale fire test. Where required elsewhere in Section 1207, large-scale fire testing shall be conducted on a representative ESS in accordance with UL 9540A. The testing shall be conducted or witnessed and reported by an *approved* testing laboratory and show that a fire involving one ESS will not propagate to an adjacent ESS, and where installed within buildings, enclosed areas and walk-in units will be contained within the room, enclosed area or walk-in unit for the duration of the test. The test report shall be provided to the *fire code official* for review and approval in accordance with Section 104.2.2. (Material based on NFPA 855 2023 Ed.)

C This section provides the details required if testing of new and innovative approaches to ESS are used. This is based on large-scale fire and fault condition testing with specific criteria for success and failure. The UL 9540A Test Method was specifically developed to cover this testing. Since extinguishment of a fire involving a thermal runaway in an ESS is extremely difficult, if not impossible, the code requirements are based on the concept of an ESS unit being completely consumed by fire without propagating to adjacent ESS. A large-scale fire test is essential to documenting that unit-to-unit propagation will not occur as well as generating additional data for necessary risk assessments for varied installation locations and arrangements relative to exposures. If the UL 9540A test results did not include a large-scale fire where the ESS unit is completely consumed by fire, a separate large-scale fire test must be conducted and test data accompanied by a detailed analysis is then supplied to the fire code official.

1207.1.8 Fire remediation. Where a fire or other event has damaged the ESS and ignition or re-ignition of the ESS is possible, the system *owner*, agent or lessee shall take the following actions, at their expense, to mitigate the hazard or remove damaged equipment from the premises to a safe location.

C This section addresses fire events involving technologies such as lithium-ion battery systems, since lithium-ion battery fires have the potential to reignite hours or even days after initial extinguishment by the fire department, which cannot remain on the scene indefinitely until the fire-damaged ESS is safely removed from the premises. The fire remediation requirements, similar to fire watch requirements, make the owner responsible for sending mitigation personnel to the scene to take over the remediation process.

1207.1.8.1 Fire mitigation personnel. Where, in the opinion of the *fire code official*, it is essential for public safety that trained personnel be on-site to respond to possible ignition or re-ignition of a damaged ESS, the system owner, agent or lessee shall dispatch within 15 minutes one or more fire mitigation personnel to the premise, as required and *approved*, at their expense. These personnel shall remain on duty continuously after the fire department leaves the premise until the damaged energy storage equipment is removed from the premises, or earlier if the *fire code official* indicates the public safety hazard has been abated. (Material based on NFPA 855 2023 Ed.)

C This section sets up the personnel in advance so that if fire mitigation is required beyond the initial event, there are qualified personnel ready to be dispatched. It is important that the assigned personnel have the requisite level of knowledge to recognize and effectively respond to the specific hazards of the damaged ESS. This section clarifies who is responsible for providing this service. It also sets out that the owner is financially responsible for such services.

1207.1.8.2 Duties. On-duty fire mitigation personnel shall have the following responsibilities:

1. Keep a diligent watch for fires, obstructions to *means of egress* and other hazards.
2. Immediately contact the fire department if their assistance is needed to mitigate any hazards or extinguish fires.

3. Take prompt measures for remediation of hazards in accordance with the decommissioning plan per Section 1207.2.3.
4. Take prompt measures to assist in the evacuation of the public from the structures. (Material based on NFPA 855 2023 Ed.)

C This section simply describes the specific responsibilities of such fire mitigation personnel. This is similar to Section 403.11.1.2 for fire watch personnel.

1207.2 Commissioning, decommissioning, operation and maintenance. Commissioning, decommissioning, operation and maintenance shall be conducted in accordance with this section.

C This section covers commissioning, decommissioning, maintenance and testing requirements, which are important considerations for providing a safe, code-compliant installation and long-term successful performance.

1207.2.1 Commissioning. Commissioning of newly installed ESS and existing ESS that have been retrofitted, replaced or previously decommissioned and are returning to service shall be conducted prior to the ESS being placed in service in accordance with a commissioning plan that has been *approved* prior to initiating commissioning. The commissioning plan shall include the following:

1. A narrative description of the activities that will be accomplished during each phase of commissioning, including the personnel intended to accomplish each of the activities.
2. A listing of the specific ESS and associated components, controls and safety-related devices to be tested, a description of the tests to be performed and the functions to be tested.
3. Conditions under which all testing will be performed, which are representative of the conditions during normal operation of the system.
4. Documentation of the owner's project requirements and the basis of design necessary to understand the installation and operation of the ESS.
5. Verification that required equipment and systems are installed in accordance with the *approved* plans and specifications.
6. Integrated testing for all fire and safety systems.
7. Testing for any required thermal management, ventilation or exhaust systems associated with the ESS installation.
8. Preparation and delivery of operation and maintenance documentation.
9. Training of facility operating and maintenance staff.
10. Identification and documentation of the requirements for maintaining system performance to meet the original design intent during the operation phase.
11. Identification and documentation of personnel who are qualified to service, maintain and decommission the ESS, and respond to incidents involving the ESS, including documentation that such service has been contracted for.
12. A decommissioning plan for removing the ESS from service, and from the facility in which it is located. The plan shall include details on providing a safe, orderly shutdown of energy storage and safety systems with notification to the code officials prior to the actual decommissioning of the system. The decommissioning plan shall include contingencies for removing an intact operational ESS from service, and for removing an ESS from service that has been damaged by a fire or other event.

Exceptions:

1. Commissioning shall not be required for lead-acid and nickel-cadmium battery systems at facilities under the exclusive control of communications utilities that comply with NFPA 76 and operate at less than 50 VAC and 60 VDC. A decommissioning plan shall be provided and maintained where required by the *fire code official*.
2. Lead-acid and nickel-cadmium battery systems less than 50 VAC, 60 VDC that are in telecommunications facilities for installations of communications equipment under the exclusive control of communications utilities, and are located outdoors or in building spaces or walk-in units used exclusively for such installations that are in compliance with NFPA 76, shall be permitted to have a commissioning plan in compliance with recognized industry practices in lieu of complying with Section 1207.2.1.
3. Lead-acid and nickel-cadmium battery systems that are used for DC power for control of substations and control or safe shutdown of generating stations under the exclusive control of the electric utilities, and are located in building spaces or walk-in units used exclusively for such installations, shall be permitted to have a commissioning plan in compliance with applicable governmental laws and regulations in lieu of developing a commissioning plan in accordance with Section 1207.2.1. (Material based on NFPA 855 2023 Ed.)

C The main intent of this section is to make sure that an ESS installation will perform as designed. The requirements of this section are best practice requirements that communicate how the system is to work and verify that it is in fact operating as designed. This process inherently reduces the risk of system failure. This includes the need to provide a plan for decommissioning from the very start to properly address the lifecycle of the system.

Several proven technologies are exempted from the need for commissioning plans. The first exception recognizes facilities under the exclusive control of communications utilities and that meet NFPA 76 do not have to provide a commissioning plan if they use lead-acid and nickel-cadmium battery systems less than 50 volts alternating current (VAC) and 60 volts direct current (VDC). This exception does require as a condition that a decommissioning plan be provided to the fire code official.

Exception 2 allows those same communications utilities that have outdoor, indoor or walk-in units under their exclusive use to have a commissioning plan that meets recognized industry practices in lieu of complying with the commissioning requirements required by Section 1207.

Exception 3 does not require compliance with the commissioning requirements in Section 1207 for electric utilities using lead-acid and nickel-cadmium battery systems for DC power for substation control and generating stations control or safe shutdown that meet applicable governmental laws and regulations.

1207.2.1.1 Initial acceptance testing. During the commissioning process an ESS shall be evaluated for proper operation in accordance with the manufacturer's instructions and the commissioning plan prior to final approval.

C This section ensures that the verification process occurs and that such process is consistent with both the manufacturer's instructions and established commissioning plan. ESS commissioning involves "cold" commissioning, which does not involve charging or cycling the batteries, and "hot" commissioning, which involves energizing the systems to charge and cycle the batteries. It is important that all fire protection system acceptance tests be completed before "hot" commissioning occurs.

1207.2.1.2 Commissioning report. A report describing the results of the system commissioning, including the results of the initial acceptance testing required in Section 1207.2.1.1, shall be provided to the *fire code official* prior to final inspection and approval and maintained at an *approved* on-site location. (Material based on NFPA 855 2023 Ed.)

C This section requires that the commissioning process is documented prior to final approval. This report may be critical in the future and needs to be placed in an approved on-site location. The location of the commissioning report will vary and will be specific to each installation. It should not be located in potentially hazardous locations but must be convenient to the ESS installation. There is a possibility it could be electronically available; however, this section does not specifically speak to this issue.

1207.2.2 Operation and maintenance. An operation and maintenance manual shall be provided to both the ESS *owner* or their authorized agent and the ESS operator before the ESS is put into operation and shall include the following:

1. Manufacturer's operation manuals and maintenance manuals for the entire ESS, or for each component of the system requiring maintenance, that clearly identify the required routine maintenance actions.
2. Name, address and phone number of a service agency that has been contracted to service the ESS and its associated safety systems.
3. Maintenance and calibration information, including wiring diagrams, control drawings, schematics, system programming instructions and control sequence descriptions, for all energy storage control systems.
4. Desired or field-determined control set points that are permanently recorded on control drawings at control devices or, for digital control systems, in system programming instructions.
5. A schedule for inspecting and recalibrating all ESS controls.
6. A service record log form that lists the schedule for all required servicing and maintenance actions and space for logging such actions that are completed over time and retained on-site.

The ESS shall be operated and maintained in accordance with the manual and a copy of the manual shall be retained at an approved on-site location.

C This manual is part of the package of requirements that ensures proper operation and maintenance to reduce the likelihood of a failure. All the pertinent information is contained in this manual for ongoing operation, including the contact information for those who undertake maintenance or repairs on the system. The schedule for inspecting and recalibrating ESS controls is included along with a service log to ensure the system is properly maintained. Similar to the commissioning report, a copy of the manual needs to be located in an approved on-site location. The scheduling of inspections for fire protection equipment outlined in NFPA 25, NFPA 72, NFPA 69 and other referenced standards should be included in this manual.

1207.2.2.1 Ongoing inspection and testing. Systems that monitor and protect the ESS installation shall be inspected and tested in accordance with the manufacturer's instructions and the operation and maintenance manual. Inspection and testing records shall be maintained in the operation and maintenance manual.

C In addition to the ESS itself, systems such as gas detection, ventilation, smoke detection and automatic sprinklers also need to be maintained to assist in mitigating any failures. These records also need to be included in the operation and maintenance manual.

1207.2.3 Decommissioning. The code official shall be notified prior to the decommissioning of an ESS. Decommissioning shall be performed in accordance with the decommissioning plan that includes the following:

1. A narrative description of the activities to be accomplished for removing the ESS from service, and from the facility in which it is located.
2. A listing of any contingencies for removing an intact operational ESS from service, and for removing an ESS from service that has been damaged by a fire or other event. (Material based on NFPA 855 2023 Ed.)

C Decommissioning is as critical as commissioning when it comes to ESS. Such systems are often subject to thermal runaway even when they are no longer in use. Therefore, they must be properly removed from service, which typically means

removal from the facility. The requirements will vary based on whether it is a functioning system or one that is damaged. Though the commissioning report is required to contain a decommissioning report, it is recognized that if decommissioning is due to damage from fire or other event, the specific details of safe decommissioning will be incident specific.

1207.3 Equipment. ESS equipment shall be in accordance with Sections 1207.3.1 through 1207.3.9.

C This section is focused on the requirements for ESS equipment, which includes listing requirements and requirements related to repairs, retrofits and replacements when systems need to be upgraded or serviced.

1207.3.1 Energy storage system listings. ESS shall be *listed* in accordance with UL 9540.

Exceptions:

1. Lead-acid and nickel-cadmium battery systems less than 50 VAC, 60 VDC in telecommunications facilities for installations of communications equipment under the exclusive control of communications utilities located outdoors or in building spaces used exclusively for such installations that are in compliance with NFPA 76.
2. Lead-acid and nickel-cadmium battery systems that are used for DC power for control of substations and control or safe shutdown of generating stations under the exclusive control of the electric utility, and located outdoors or in building spaces used exclusively for such installations.
3. Lead-acid battery systems in uninterruptible power supplies listed and *labeled* in accordance with UL 1778 and utilized for standby power applications.

(Material based on NFPA 855 2023 Ed.)

C Energy storage systems are required to be listed in accordance with UL 9540. This standard covers the listing of storage batteries and all the equipment associated with the system, including cabinets and prefabricated enclosures. There are three exceptions focused on lead-acid and nickel-cadmium battery technologies and applications with a proven safety record and longstanding compliance paths. All three focus on Lead-acid and nickel-cadmium battery technologies.

1207.3.2 Equipment listing. Chargers, inverters and energy storage management systems shall be covered as part of the UL 9540 listing or shall be *listed* separately.

C There are sometimes separate listings that address chargers, inverters and energy management systems where they are not part of the UL 9540 listing. If they are not part of the UL 9540 listing, they must be addressed individually by specific listing standards.

1207.3.3 Utility interactive systems. Inverters shall be *listed* and *labeled* in accordance with UL 1741. Only inverters *listed* and *labeled* for utility interactive system use and identified as interactive shall be allowed to operate in parallel with the electric utility power system to supply power to common loads.

C This section requires a listing in accordance with UL 1741 where it is not part of the UL 9540 listing. An inverter is an electronic device that is capable of changing direct current (DC) to alternating current (AC). Inverters need to be specifically listed and labeled to avoid using the wrong type and subsequently leading to system failure. Note that this section also requires a specific listing for utility interactive systems. These types of inverters allow things like automatic transfer to enable the ESS to feed power back into the commercial power grid or disconnect from the commercial grid when it is supposed to be de-energized due to an outage.

1207.3.4 Energy storage management system. Where required by the ESS listing, an *approved* energy storage management system that monitors and balances cell voltages, currents and temperatures within the manufacturer's specifications shall be provided. The system shall disconnect electrical connections to the ESS or otherwise place it in a safe condition if potentially hazardous temperatures or other conditions such as short circuits, over voltage or under voltage are detected. (Material based on NFPA 855 2023 Ed.)

C An energy management system is an important safety system that monitors and takes actions on abnormal conditions that could lead to failures. These systems are critical to reducing a risk of failure.

1207.3.5 Enclosures. Enclosures of ESS shall be of noncombustible construction. (Material based on NFPA 855 2023 Ed.)

C This prevents other methods of fire spread from one ESS to another or other combustibles within a building.

1207.3.6 Repairs. Repairs of ESS shall only be done by qualified personnel. Repairs with other than identical parts shall be considered retrofitting and comply with Section 1207.3.7. Repairs shall be documented in the service records log. (Material based on NFPA 855 2023 Ed.)

C Several requirements are addressed in this section. The first is the necessary qualifications of those undertaking the repairs. This is especially important given that these systems are complex and at risk of failure if the repairs are done incorrectly. Second, system innovations are happening quickly, and identical parts may not be available. To ensure compatibility in those cases, the replacement parts and equipment need to be addressed as a retrofit in accordance with Section 1207.3.7. This section requires all repairs to be documented.

1207.3.7 Retrofits. Retrofitting of an existing ESS shall comply with the following:

1. A construction permit shall be obtained in accordance with Section 105.6.6.
2. New batteries, battery modules, capacitors and similar ESS components shall be *listed*.
3. Battery management and other monitoring systems shall be connected and installed in accordance with the manufacturer's instructions.
4. The overall installation shall continue to comply with UL 9540 listing requirements, where applicable.
5. Systems that have been retrofitted shall be commissioned in accordance with Section 1207.2.1.
6. Retrofits shall be documented in the service records log. (Material based on NFPA 855 2023 Ed.)

C This section addresses practices to be followed when systems need to be upgraded or serviced. These systems are very sensitive to changes that could lead to thermal runaway, so clear requirements for retrofit need to be provided. This includes obtaining a construction permit to notify the fire department a retrofit is occurring. In addition, batteries must be properly listed and the overall installation must comply with UL 9540. Finally, the system needs to be commissioned to demonstrate performance, and records of the process need to be provided.

1207.3.7.1 Retrofitting lead acid and nickel cadmium. Changing out or retrofitting of lead-acid and nickel-cadmium batteries with other lead-acid and nickel-cadmium batteries in the following applications shall be considered repairs where there is no increase in system size or energy capacity greater than 10 percent of the original design.

1. At facilities under the exclusive control of communications utilities that comply with NFPA 76 and operate at less than 50 VAC and 60 VDC.
2. Battery systems used for DC power for control of substations and control or safe shutdown of generating stations under the exclusive control of the electric utility, and located outdoors or in building spaces used exclusively for such installations.
3. Batteries in uninterruptible power supplies *listed* and *labeled* in accordance with UL 1778 and used for standby power applications only.

(Material based on NFPA 855 2023 Ed.)

C This section provides some relief to operations that change out or retrofit lead-acid and nickel-cadmium batteries where there is no increase in system size or energy capacity greater than 10 percent of the original design. This section outlines the type of applications that can be treated as "repairs" and do not need operational or construction permits.

1207.3.8 Replacements. Replacements of ESS shall be considered new ESS installations and shall comply with the provisions of Section 1207 as applicable to new ESS. The ESS being replaced shall be decommissioned in accordance with Section 1207.2.3. (Material based on NFPA 855 2023 Ed.)

C This section requires that replacement of ESS shall be addressed no differently than as new and Section 1207 would apply to this installation. Where an ESS system is being replaced, it is important that such systems are properly decommissioned since the existing system may still be energized or subject to thermal runaway.

1207.3.9 Reused and repurposed equipment. Equipment and materials shall only be reused or reinstalled as permitted in Section 104.9.1. Storage batteries previously used in other applications, such as electric vehicle propulsion, shall not be reused in applications regulated by Chapter 12 unless *approved* by the *fire code official* and unless the equipment is refurbished by a battery refurbishing company *approved* in accordance with UL 1974. (Material based on NFPA 855 2023 Ed.)

C This section allows fire code officials to regulate installations of repurposed electric vehicle batteries that are converted for ESS use in buildings. It is important to recognize that UL 1974 applies to the process used by the firm refurbishing the battery and is not a listing of the battery.

1207.4 General installations requirements. Stationary and mobile ESS shall comply with the requirements of Sections 1207.4.1 through 1207.4.12.

C This section is focused on a number of requirements that facilitate the safe operation of ESS such as proper electrical disconnects, clearances, seismic supports and separation from combustibles. These features are part of the package of safety measures that reduce the risk of a failure.

1207.4.1 Electrical disconnects. Where the ESS disconnecting means is not within sight of the main electrical service disconnecting means, placards or directories shall be installed at the location of the main electrical service disconnecting means indicating the location of stationary storage battery system disconnecting means in accordance with NFPA 70.

Exception: Electrical disconnects for lead-acid and nickel-cadmium battery systems at facilities under the exclusive control of communications utilities and operating at less than 50 VAC and 60 VDC shall be permitted to have electrical disconnects signage in accordance with NFPA 76.

C ESS may be in various locations within buildings and may not be near the main service disconnect. Therefore, a way of communicating the location of the disconnect for such systems is necessary at the main service disconnect. This is critical to fighting a fire and being able to disconnect all power supplies affecting firefighting operations. Note that to ensure reliability of the telecommunications network, batteries utilized in telecommunications power plants in central offices rarely

have circuit breakers or an automatic disconnecting means in order to satisfy an FCC Best Practice designed to ensure the highest reliability. Such installations are not within the scope of NFPA 70. In this case, signage is required in compliance with NFPA 76, which is more specific to telecommunications facilities.

1207.4.2 Working clearances. Access and working space shall be provided and maintained about all electrical equipment to permit ready and safe operation and maintenance of such equipment in accordance with NFPA 70 and the manufacturer's instructions.

C This section requires the proper spacing to reduce ignition of surrounding materials and to allow proper maintenance. This is consistent with Section 603.4.

1207.4.3 Fire-resistance-rated separations. Rooms and other indoor areas containing ESS shall be separated from other areas of the building in accordance with Section 1207.7.4. ESS shall be permitted to be in the same room with the equipment they support.

C This section focuses on containment of the hazard to the location where the ESS is located, whether in an entire room or in a small space such as a closet. ESS being subject to thermal runaway could cause a fire that will continue to burn for an extended amount of time. Section 1207.7.4 requires 2-hour fire barriers and 2-hour horizontal separations. In many cases, due to the impracticality of separating ESS from the equipment being supported, it is clarified that they may remain together in the same room.

1207.4.4 Seismic and structural design. Stationary ESS shall comply with the seismic design requirements in Chapter 16 of the *International Building Code*, and shall not exceed the floor loading limitation of the building.

C Some of these installations are quite heavy and can potentially affect the structural integrity of the building. Therefore, the floor loading limitations need to be addressed. Also, to avoid additional failures during seismic events, the requirements of the IBC must be addressed. Assessment of rooftop installations on existing buildings must include whether structural fire resistance of the primary structural frame and bearing walls will remain in compliance with IBC Table 601.

1207.4.5 Vehicle impact protection. Where ESS are subject to impact by a motor vehicle, including forklifts, vehicle impact protection shall be provided in accordance with Section 312.

C This section provides the requirements for protecting the battery system from physical damage to avoid a malfunction of the systems and the hazards that could result. This applies to indoor or outdoor installations, depending on the potential impact risk from motor vehicles (see commentary, Section 312). Additional concepts of impact protection can be found in Section 1207.11.7.1.

1207.4.6 Combustible storage. Combustible materials shall not be stored in ESS rooms, areas or walk-in units. Combustible materials in occupied work centers covered by Section 1207.4.10 shall be stored at least 3 feet (914 mm) from ESS cabinets.

C In the past, ESS were often independent of the rest of the building and its operations. In particular, the focus had only been on applications such as telecommunications buildings. However, ESS are now found in a variety of locations with other ongoing activities, such as data centers. This section requires a minimum separation from combustibles or prohibits the storage of combustible materials to reduce the risk of fire.

1207.4.7 Toxic and highly toxic gases. ESS that have the potential to release toxic and highly toxic gas during charging, discharging and normal use conditions shall be provided with a hazardous exhaust system in accordance with Section 502.8 of the *International Mechanical Code*.

C This section covers current and potential battery technologies that could produce toxic gases during charging, discharging and normal use, triggering Chapter 60 safety requirements. There is no intent to address batteries that produce toxic gases during fires or abnormal conditions. Hydrogen and oxygen produced by vented batteries are not considered toxic or highly toxic gases.

1207.4.8 Signage. *Approved* signs shall be provided on or adjacent to all entry doors for ESS rooms or areas and on enclosures of ESS cabinets and walk-in units located outdoors, on rooftops or in open parking garages. Signs designed to meet both the requirements of this section and NFPA 70 shall be permitted. The signage shall include the following or equivalent:

1. "ENERGY STORAGE SYSTEM," "BATTERY STORAGE SYSTEM," "CAPACITOR ENERGY STORAGE SYSTEM" or the equivalent.
2. The identification of the electrochemical ESS technology present.
3. "ENERGIZED ELECTRICAL CIRCUITS."
4. Where water-reactive electrochemical ESS are present, the signage shall include "APPLY NO WATER."
5. Current contact information, including phone number, for personnel authorized to service the equipment and for fire mitigation personnel required by Section 1207.1.8.1.

Exception: Existing electrochemical ESS shall be permitted to include the signage required at the time they were installed. (Material based on NFPA 855 2023 Ed.)

C ESS are found in varying applications in today's world. They are no longer simply for standby and emergency power. In addition, the technologies are varied and evolving.

This section identifies the need for signage to identify the hazards. Items 1 and 2 require signage to denote where ESS are contained in a room and the type of technology present. Item 3 focuses on communicating through signage that energized electrical circuits are present.

Item 4 provides necessary information to occupants and emergency responders of any water-reactive properties and clearly communicates that water cannot be applied.

Item 5 addresses the need to provide contact information regarding personnel authorized to service the ESS and more importantly, fire mitigation personnel who are properly trained to manage the hazards associated with the ESS. See Section 1207.1.8.1.

The exception addresses currently installed systems that were not required to have the prescribed signage at the time of installation, which often will be stand-alone lead-acid battery systems. These types of installations have had a good safety record and typically do not create exposure hazards based on their chemistries and the way they are used and regulated.

1207.4.9 Security of installations. Rooms, areas and walk-in units in which electrochemical ESS are located shall be secured against unauthorized entry and safeguarded in an *approved* manner. Security barriers, fences, landscaping and other enclosures shall not inhibit the required air flow to or exhaust from the electrochemical ESS and its components. (Material based on NFPA 855 2023 Ed.)

C This section is general to all ESS installation types, whether indoors or outdoors. Properly securing such installations reduces the risk of arson or vandalism, which could lead to a failure of such systems. This is more of a concern with outdoor installations or walk-in units. It is important that the security features do not negatively impact the ventilation requirements for the installation. Facilities with a secured perimeter and controlled access to the site would qualify, and additional securing of an ESS installation on such a site would not be required.

1207.4.10 Occupied work centers. Electrochemical ESS located in rooms or areas occupied by personnel not directly involved with maintenance, service and testing of the systems shall comply with the following:

1. Electrochemical ESS located in occupied work centers shall be housed in locked noncombustible cabinets or other enclosures to prevent access by unauthorized personnel.
2. Where electrochemical ESS are contained in cabinets in occupied work centers, the cabinets shall be located within 10 feet (3048 mm) of the equipment that they support.
3. Cabinets shall include signage complying with Section 1207.4.8. (Material based on NFPA 855 2023 Ed.)

C In cases where the ESS is located where building occupants could be present, some level of protection is necessary.

This section first requires the ESS to be secured. Similar to Section 1207.4.9, this reduces a possible failure of the system due to vandalism or arson.

Placing ESS cabinets within 10 feet (3048 mm) of the equipment served by the system will limit the number of devices that can be powered by the system due to the voltage drop associated with longer or multiple conductor runs. This section also enhances the safety of personnel who may have to secure and work on equipment served by the ESS by placing the secondary power supply near enough to the powered equipment to ensure line-of-sight supervision of the secured ESS during equipment repair or maintenance.

This section calls for the identification of the specific hazards associated with the ESS contained in each cabinet. ESS technologies vary and with that, the hazard varies. This information will also provide the capacity of each system, which is essential for the safety of the occupants and first responders.

1207.4.11 Open rack installations. Where electrochemical ESS are installed in a separate equipment room and only authorized personnel have access to the room, they shall be permitted to be installed on an open rack for ease of maintenance. (Material based on NFPA 855 2023 Ed.)

C This section essentially allows ESS to be on open racks where access to the room is restricted. The purpose is to facilitate maintenance, which could have the effect of better safety performance. Essentially, the ESS utilizes the room as a means of separation in place of an enclosure.

1207.4.12 Walk-in units. Walk-in units shall be entered only for inspection, maintenance and repair of ESS units and ancillary equipment, and shall not be occupied for other purposes.

C This section clarifies the intent of walk-in units. Such units are not intended for regular occupancy and are only to be entered for the purposes of maintenance and repairs. Those entering are trained to do so. Although occupancy is restricted, ESS walk-in units are defined as buildings, and all requirements applicable to buildings apply, including fire suppression.

1207.5 Electrochemical ESS protection. The protection of electrochemical ESS shall be in accordance with Sections 1207.5.1 through 1207.5.8 where required by Sections 1207.7 through 1207.10. (Material based on NFPA 855 2023 Ed.)

C This section generally sets out the specific protection required for the ESS. Section 1207.4 focuses on the elements surrounding ESS, but not the specific protection of the equipment. The triggers for these protection features vary based on the

technology used and the type of installation addressed in Sections 1207.7 through 1207.10. Table 1207.5 is intended to trigger when the provisions of Section 1207.5 become applicable.

TABLE 1207.5—MAXIMUM ALLOWABLE QUANTITIES OF ELECTROCHEMICAL ESS	
TECHNOLOGY	MAXIMUM ALLOWABLE QUANTITIES ^a
STORAGE BATTERIES	
Flow batteries ^b	600 kWh
Lead-acid, all types	Unlimited
Lithium-ion	600 kWh
Nickel-cadmium (Ni-Cd), nickel-metal hydride (NI-MH) and nickel zinc (Ni-Zn)	Unlimited
Sodium nickel chloride	600 kWh
Zinc-manganese dioxide (Zn-MnO ₂)	Unlimited
Other battery technologies	200 kWh
CAPACITORS	
All types	20 kWh
OTHER ELECTROCHEMICAL ESS	
All types	20 kWh
For SI: 1 kilowatt hour = 3.6 megajoules.	
a. For electrochemical ESS units rated in amp-hours, kWh shall equal rated voltage times the amp-hour rating divided by 1,000.	
b. Shall include vanadium, zinc-bromine, polysulfide-bromide and other flowing electrolyte-type technologies.	

C This table sets out the MAQ of each technology that can be found in a single installation within rooms, areas and walk-in units. Where these MAQs are exceeded, Section 1207.5.2 would allow such installations under certain conditions based on proven technologies such as lead acid and nickel cadmium at communication utilities, the study of the hazard through full-scale fire tests or by locating within dedicated-use buildings. The exceptions found in Section 1207.5.2 do not exempt such technologies from protection; it is simply the size of the installation that is not limited.

1207.5.1 Size and separation. Electrochemical ESS shall be segregated into groups not exceeding 50 kWh (180 megajoules). Each group shall be separated a minimum of 3 feet (914 mm) from other groups and from walls in the storage room or area. The storage arrangements shall comply with Chapter 10.

Exceptions: (Material based on NFPA 855 2023 Ed.)

1. Lead-acid and nickel-cadmium battery systems in facilities under the exclusive control of communications utilities and operating at less than 50 VAC and 60 VDC in accordance with NFPA 76.
2. Lead-acid and nickel-cadmium systems that are used for DC power for control of substations and control or safe shut-down of generating stations under the exclusive control of the electric utility, and located outdoors or in building spaces used exclusively for such installations.
3. Lead-acid battery systems in uninterruptible power supplies *listed* and *labeled* in accordance with UL 1778, utilized for standby power applications, and limited to not more than 10 percent of the floor area on the floor on which the ESS is located.
4. The *fire code official* is authorized to approve larger capacities or smaller separation distances based on large-scale fire testing complying with Section 1207.1.5.

C To reduce the possible size and intensity of a fire or explosion, ESS units are limited in capacity and must be separated into certain capacity groups. Each group must have a fire break of not less than 3 feet from other groups and walls.

Exceptions 1 through 3 acknowledge the safety record of lead-acid and nickel-cadmium batteries and do not require the 50 kWh and 3-foot separation.

Exception 4 provides flexibility to the base capacity of the ESS group if testing determines that a fire would be contained to the room as separated. It should also be noted that the types of installation that are addressed in Sections 1207.7 through 1207.10 will affect whether these size and separation requirements apply. For example, Table 1207.8 for remote outdoor installations would not require compliance with this section.

1207.5.2 Maximum allowable quantities. *Fire areas* within rooms, areas and walk-in units containing electrochemical ESS shall not exceed the maximum allowable quantities in Table 1207.5.

Exceptions: (Material based on NFPA 855 2023 Ed.)

1. Where *approved* by the *fire code official*, rooms, areas and walk-in units containing electrochemical ESS that exceed the amounts in Table 1207.5 shall be permitted based on a hazardous mitigation analysis in accordance with Section 1207.1.6 and large-scale fire testing complying with Section 1207.1.7.

2. Lead-acid and nickel-cadmium battery systems installed in facilities under the exclusive control of communications utilities, and operating at less than 50 VAC and 60 VDC in accordance with NFPA 76.
3. Dedicated-use buildings in compliance with Section 1207.7.1.

C Scientific research and large-scale fire and fault condition testing historically are not available to justify allowing unlimited quantities of energy in an ESS to be provided in a room, area or walk-in unit. By referencing Table 1207.5, this section establishes a maximum 600 kWh MAQ for each fire area and 200 kWh for certain technologies for which less information is available and in effect serves as a trigger for large-scale fire testing. The MAQ of 600 kWh is equivalent to 1,250 storage batteries, each rated 12 V, 40 Ah. In addition, capacitors and other electrochemical ESS are limited to 20 kWh.

There are several exceptions to these requirements. Exception 1 allows such installations to increase capacity where a successful hazards mitigation analysis has been completed. This would involve larger-scale fire testing of the installation.

Exception 2 recognizes the safety record of lead-acid and nickel-cadmium battery systems under the control of communications utilities.

Exception 3 recognizes the minimal life safety hazard associated with dedicated-use buildings that comply with Section 1207.7.1.

1207.5.2.1 Mixed electrochemical energy systems. Where rooms, areas and walk-in units contain different types of electrochemical energy technologies, the total aggregate quantities of the systems shall be determined based on the sum of percentages of each technology-type quantity divided by the maximum allowable quantity of each technology type. The sum of the percentages shall not exceed 100 percent of the maximum allowable quantity.

C Because of the changing technologies of ESS, it is becoming more common to have multiple technologies in a single building. This section addresses how to determine the MAQ large-scale fire test trigger where multiple ESS are used. Note that due to the safety record of lead-acid and nickel-cadmium battery systems, the MAQ are unlimited. If, for example, there are flow batteries and sodium batteries in the same fire area, the allowable installed quantity permitted under this section of the code is determined on a percentage basis as follows:

Given:

450 kWh flow battery system

75 kWh other battery system

450 kWh flow battery system = $450 \text{ kWh} / 600 \text{ kWh} = 0.75 \times 100 = 75\%$

75 kWh other battery technology system = $75 \text{ kWh} / 200 \text{ kWh} = 0.375 \times 100 = 37.5\%$

75% (flow) + 37.5% (other) = 112.5% of MAQ; Therefore, hazard mitigation analysis would be required for other than dedicated-use buildings.

1207.5.3 Elevation. Electrochemical ESS shall not be located in the following areas:

1. Where the floor is located more than 75 feet (22 860 mm) above the lowest level of fire department vehicle access.
2. Where the floor is located below the lowest *level of exit discharge*.

Exceptions: (Material based on NFPA 855 2023 Ed.)

1. Lead-acid and nickel-cadmium battery systems less than 50 VAC and 60 VDC installed in facilities under the exclusive control of communications utilities in accordance with NFPA 76.
2. Lead-acid and nickel-cadmium systems that are used for DC power for control of substations and control or safe shutdown of generating stations under the exclusive control of the electric utility, and located outdoors or in building spaces used exclusively for such installations.
3. Lead-acid battery systems in uninterruptable power supplies *listed and labeled* in accordance with UL 1778, utilized for standby power applications, which is limited to not more than 10 percent of the floor area on the floor on which the ESS is located.
4. Where *approved*, installations shall be permitted in underground vaults complying with NFPA 70, Article 450, Part III.
5. Where *approved by the fire code official*, installations shall be permitted on higher and lower floors.

C In terms of location, the main concerns are access for firefighting and potential intensity of fires from certain ESS technologies such as lithium-ion. ESS are becoming more common in high-rise buildings and without this section, such installations can make it much more difficult to fight a fire. This section does not prohibit installations in high-rise buildings but instead limits the location of such systems to the low-rise portion of the building. A key component for fire attack is the amount of available water, and 75 feet was chosen based on a fire department's ability to use pre-piped elevating tower ladders or ladders to bring a larger water supply to upper floors. Likewise, underground buildings also are addressed. Fires involving ESS can generate tremendous amounts of smoke and heat, which firefighters would have to climb down into for floors lower than the level of exit discharge. These restrictions were specifically added based on concerns voiced by the operations side of the fire service.

Exception 1 is provided for lead-acid and nickel-cadmium storage battery systems, as they have a good safety record and further restrictions beyond that provided in previous editions of this code are not necessary. Exceptions 2 and 3 align the requirements with industry practices and other code sections. Exception 4 allows installations in underground vaults that comply with NFPA 70 that essentially provides 3 hours of fire-resistance-rated construction. Exception 5 gives some flexibility to unique situations where the risks to such installations are properly mitigated and such installations are specifically approved. The amount of water available for firefighting to floors over 75 feet (22 860 mm) above the lowest level of fire department vehicle access would be a consideration for installations on higher floors. The ability to control and exhaust heat and smoke from lower floors would be a consideration for allowing installation on lower floors.

1207.5.4 Fire detection. An *approved automatic smoke detection system* or radiant energy-sensing fire detection system complying with Section 907.2 shall be installed in rooms, indoor areas and walk-in units containing electrochemical ESS. An *approved* radiant energy-sensing fire detection system shall be installed to protect open parking garage and rooftop installations. Alarm signals from detection systems shall be transmitted to a central station, proprietary or remote station service in accordance with NFPA 72, or where *approved* to a constantly attended location.

Exception: Normally unoccupied, remote stand-alone telecommunications structures with a gross floor area of less than 1,500 square feet (139 m²) utilizing lead-acid or nickel-cadmium batteries shall not be required to have a fire detection system installed. (Material based on NFPA 855 2023 Ed.)

C Many ESS fires are likely to be slow growing and smoldering; therefore, a smoke detection system or radiant energy-sensing fire detection system is required for early detection and warning in accordance with Section 907.2 where the system requires application of Section 1207 (i.e., exceeds the threshold quantities in Table 1207.1.3). All types of installations addressed by Sections 1207.7 through 1207.10 require fire detection. For example, where an ESS using lithium-ion batteries exceeds 20 kWh, a smoke detection or radiant energy-sensing system must be provided. The goal is early warning of an event to assist with response for the fire department. For radiant-energy detection, that would equate to thermal imaging or multispectrum infrared flame detectors, both of which are recognized by NFPA 72.

The exception addresses small remote telecommunications facilities that normally are not occupied. These structures often accompany cellular telephone towers in remote mountainous locations that are separated from the wildland by open space. Fire detection systems are intended to detect and report fires to the local fire department. These remote areas might not have a fire department, or if they do, fire may destroy these structures before the fire department arrives.

1207.5.4.1 System status. Lead-acid and nickel-cadmium battery systems that are used for DC power for control of substations and control or safe shutdown of generating stations under the exclusive control of the electric utility, and located outdoors or in building spaces used exclusively for such installations, shall be allowed to use the process control system to monitor the smoke or radiant energy-sensing fire detectors required in Section 1207.5.4. (Material based on NFPA 855 2023 Ed.)

C Lead-acid and nickel-cadmium battery systems that meet the criteria of this section can use the process control system to monitor the required smoke or radiant energy-sensing fire detectors instead of monitoring as required by Section 907.

1207.5.5 Fire suppression systems. Rooms and areas within buildings and walk-in units containing electrochemical ESS shall be protected by an automatic fire suppression system designed and installed in accordance with one of the following:

1. *Automatic sprinkler systems* designed and installed in accordance with Section 903.3.1.1 for ESS units (groups) with a maximum stored energy capacity of 50 kWh, as described in Section 1207.5.1, shall be designed with a minimum density of 0.3 gpm/ft² (1.14 L/min) based over the area of the room or 2,500 square-foot (232 m²) design area, whichever is smaller, unless a lower density is approved based on large-scale fire testing in accordance with Section 1207.1.7.
2. *Automatic sprinkler systems* designed and installed in accordance with Section 903.3.1.1 for ESS units (groups) exceeding 50 kWh shall use a density based on large-scale fire testing complying with Section 1207.1.7.
3. The following alternative *automatic fire-extinguishing systems* designed and installed in accordance with Section 904, provided that the installation is *approved* by the *fire code official* based on large-scale fire testing complying with Section 1207.1.7:
 - 3.1. NFPA 12, *Standard on Carbon Dioxide Extinguishing Systems*.
 - 3.2. NFPA 15, *Standard for Water Spray Fixed Systems for Fire Protection*.
 - 3.3. NFPA 750, *Standard on Water Mist Fire Protection Systems*.
 - 3.4. NFPA 2001, *Standard on Clean Agent Fire-Extinguishing Systems*.
 - 3.5. NFPA 2010, *Standard for Fixed Aerosol Fire-Extinguishing Systems*.

Exceptions:

1. Fire suppression systems for lead-acid and nickel-cadmium battery systems at facilities under the exclusive control of communications utilities that operate at less than 50 VAC and 60 VDC shall be provided where required by NFPA 76.
2. Lead-acid and nickel-cadmium systems that are used for DC power for control of substations and control or safe shutdown of generating stations under the exclusive control of the electric utility, and located outdoors or in building spaces used exclusively for such installations, shall not be required to have a fire suppression system installed.

3. Lead-acid battery systems in uninterruptable power supplies *listed and labeled* in accordance with UL 1778, utilized for standby power applications, which is limited to not more than 10 percent of the floor area on the floor on which the ESS is located, shall not be required to have a fire suppression system. (Material based on NFPA 855 2023 Ed.)

C Suppression systems are required for most types of ESS as addressed in Sections 1207.7 through 1207.10 with the exception of rooftop and open parking garage installations open to the sky or open parking garages where full-scale testing demonstrates that such protection is not required (see Section 1207.9.4). (Section 1207.5.5.1 addresses battery systems that contain water-reactive materials where water cannot be used.)

This section provides three options. Item 1 provides a prescriptive requirement for the use of automatic sprinkler systems with specific design criteria. This solution is only applicable to ESS units (groups) up to a maximum stored energy capacity of 50 kWh. It specifies the minimum sprinkler design criteria to be provided over the entire room or design area unless full-scale fire testing demonstrates a different design criterion is to be used.

Item 2 addresses fire sprinkler systems protecting ESS units (groups) with a stored energy capacity exceeding 50 kWh. In this case, such application must be demonstrated through a large-scale fire test. Item 3 allows those use of a number of different fire extinguishing systems as alternatives to an automatic sprinkler system. Where applying these systems, regardless of the maximum stored energy, it must be successfully demonstrated through a full-scale test. Section 904.3.2 requires that these systems address all hazards in the space unless there are multiple types of systems to address all of the fuel loads.

The exceptions apply only to those spaces or areas that are used exclusively for telecommunications equipment or under the control of electrical utilities. Historically, those spaces have a low incidence of fire events. Fires in telecommunications and utility equipment are difficult to start and, if started, grow slowly, thus permitting earlier detection as an alternative to suppression.

1207.5.5.1 Water-reactive systems. Electrochemical ESS that utilize water-reactive materials shall be protected by an *approved alternative automatic fire-extinguishing system* in accordance with Section 904, where the installation is *approved by the fire code official* based on large-scale fire testing complying with Section 1207.1.7.

C Some battery systems utilize enough water-reactive material in their construction that a traditional automatic sprinkler system is not an option. It should be noted that some battery systems such as lithium ion might be water reactive but can, in some cases, be suppressed with water effectively. This section requires that the system be based on full-scale fire and fault condition testing. Item 3 of Section 1207.5.5 allows the same concept based on large-scale testing.

1207.5.6 Maximum enclosure size. Outdoor walk-in units housing ESS shall not exceed 53 feet by 8 feet by 9.5 feet high (16 154 mm × 2438 mm × 2896 mm), not including bolt-on HVAC and related equipment, as *approved*. Outdoor walk-in units exceeding these limitations shall be considered indoor installations and comply with the requirements in Section 1207.7. (Material based on NFPA 855 2023 Ed.)

C A maximum enclosure size for walk-in units is established, corresponding to the largest ISO-type containers used for these installations, to provide differentiation between a walk-in unit and an inside installation.

1207.5.7 Vegetation control. Areas within 10 feet (3048 mm) on each side of outdoor ESS shall be cleared of combustible vegetation and other combustible growth. Single specimens of trees, shrubbery or cultivated ground cover such as green grass, ivy, succulents or similar plants used as ground cover shall be permitted to be exempt provided that they do not form a means of readily transmitting fire. (Material based on NFPA 855 2023 Ed.)

C This provision reduces exposure from exterior fires on the ESS. Some ground cover is acceptable as long as such ground cover does not create a means of readily transmitting fire. Essentially, well-manicured grass poses little hazard; however, it must be maintained to reduce the risk of possible fire spread to the ESS.

1207.5.8 Means of egress separation. ESS located outdoors and in open parking garages shall be separated from any *means of egress* as required by the *fire code official* to ensure safe egress under fire conditions, but in no case less than 10 feet (3048 mm).

Exception: The *fire code official* is authorized to approve a reduced separation distance if large-scale fire testing complying with Section 1207.1.7 is provided that shows that a fire involving the ESS will not adversely impact occupant egress. (Material based on NFPA 855 2023 Ed.)

C Separation from means of egress must be at least 10 feet to ensure safe egress if a fire occurs in an ESS installation. The exception allows a lesser separation where large-scale testing demonstrates that a hazard does not exist. This provides flexibility for new technologies or situations where a 10-foot separation is not possible.

1207.6 Electrochemical ESS technology-specific protection. Electrochemical ESS installations shall comply with the requirements of this section in accordance with the applicable requirements of Table 1207.6. (Material based on NFPA 855 2023 Ed.)

C This section simply introduces Table 1207.6, which provides technology-specific protection. These features include exhaust ventilation, explosion control, safety caps, spill control and neutralization, and thermal runaway. These features are based on each specific technology. For instance, lead-acid batteries contain liquid and will need spill control and neutralization

whereas lithium-ion batteries do not as they do not have a liquid to spill. However, both technologies are subject to thermal runaway and need proper controls to address this in accordance with Section 1207.6.5.

Note a relieves communications utilities regulated by NFPA 76 from having to provide explosion control for lead-acid and nickel-cadmium batteries given the capacity of the systems and the safety record of such facilities.

Note b states that unless proper documentation is provided for other and perhaps newer technologies, all of these features are required. The default is the more restrictive application of the requirements without further information about the performance of these ESS technologies.

Note c addresses the fact that spill control and neutralization for lead-acid and nickel-cadmium batteries is required only where such batteries are vented and have the ability to create a spill.

Note d acknowledges the fact that vented lead-acid batteries are not likely to go into thermal runaway.

Note e allows the use of a battery management system that was tested with the ESS to UL 1973 for management of thermal runaway hazards.

Note f exempts batteries with gelled electrolyte from spill control and neutralization as a spill is unlikely.

TABLE 1207.6—ELECTROCHEMICAL ESS TECHNOLOGY-SPECIFIC REQUIREMENTS

COMPLIANCE REQUIRED ^b		BATTERY TECHNOLOGY						OTHER ESS AND BATTERY TECHNOLOGIES ^b	CAPACITOR ESS ^b
Feature	Section	Lead-acid	Nickel cadmium (Ni-Cd), nickel-metal hydride (Ni-MH) and nickel zinc (Ni-Zn)	Zinc-manganese dioxide (Zn-MnO ₂)	Lithium-ion	Flow	Sodium nickel chloride		
Exhaust ventilation	1207.6.1	Yes	Yes	Yes	No	Yes	No	Yes	Yes
Explosion control	1207.6.3	Yes ^a	Yes ^a	Yes	Yes	No	Yes	Yes	Yes
Safety caps	1207.6.4	Yes	Yes	No	No	No	No	Yes	Yes
Spill control and neutralization	1207.6.2	Yes ^c	Yes ^c	Yes ^f	No	Yes	No	Yes	Yes
Thermal runaway	1207.6.5	Yes ^d	Yes	Yes ^e	Yes ^e	No	Yes	Yes ^e	Yes

a. Not required for lead-acid and nickel-cadmium batteries at facilities under the exclusive control of communications utilities that comply with NFPA 76 and operate at less than 50 VAC and 60 VDC.

b. Protection shall be provided unless documentation acceptable to the fire code official is provided in accordance with Section 104.2.2 that provides justification why the protection is not necessary based on the technology used.

c. Applicable to vented-type (i.e., flooded) nickel-cadmium and lead-acid batteries.

d. Not required for vented-type (i.e., flooded) batteries.

e. The thermal runaway protection is permitted to be part of a battery management system that has been evaluated with the battery as part of the evaluation to UL 1973.

f. Not required for batteries with gelled electrolyte.

C See the commentary to Section 1207.6.

1207.6.1 Exhaust ventilation. Where required by Table 1207.6 or elsewhere in this code, exhaust ventilation of rooms, areas and walk-in units containing electrochemical ESS shall be provided in accordance with the *International Mechanical Code* and Section 1207.6.1.1 or 1207.6.1.2.

C This section addresses the necessary ventilation requirements based on either the conclusion of the hazard mitigation analysis or the specific battery technology requirements. Some batteries need more ventilation than others.

As with any other building areas, ventilation of ESS areas must comply with the requirements of the *International Mechanical Code*[®] (IMC[®]). In many cases, basic ventilation based on human comfort will be sufficient to adequately ventilate battery spaces. The main concern with many types of batteries is the buildup of hydrogen within an enclosed space. Hydrogen has a wide flammability range and is the lightest element on the Periodic Table of Elements, having an atomic number of 1. The gas mixture is potentially explosive when the amount of hydrogen in the air exceeds 4 percent by volume. There are two options provided for ventilation addressed in Section 1207.6.1.1 or 1207.6.1.2. Section 1207.6.1.1 could be either mechanical or natural ventilation whereas Section 1207.6.1.2 is specifically focused on mechanical exhaust ventilation. The subsections that follow 1207.6.1.2 are focused on the support of mechanical ventilation.

1207.6.1.1 Ventilation based on LFL. The exhaust ventilation system shall be designed to limit the maximum concentration of *flammable gas* to 25 percent of the lower flammable limit (LFL) of the total volume of the room, area or walk-in unit during the worst-case event of simultaneous charging of batteries at the maximum charge rate, in accordance with nationally recognized standards.

C This section provides a performance-based approach by requiring a ventilation designed to limit the maximum concentration of flammable gas to 25 percent of the LFL of the total volume of the room, area or walk-in unit with the worst-case event of charging at the maximum charging rate.

1207.6.1.2 Ventilation based on exhaust rate. Mechanical exhaust ventilation shall be provided at a rate of not less than 1 ft³/min/ft² (5.1 L/sec/m²) of floor area of the room, area or walk-in unit. The ventilation shall be either continuous or shall be activated by a *gas detection system* in accordance with Section 1207.6.1.2.4.

C This section simply prescribes a ventilation rate of not less than 1 cubic foot per minute per square foot [1 ft³/min/ft² or 0.0051 m³/(s × m²)] of room area. This is a conservative ventilation rate. This section does not require continuous ventilation where ventilation is activated by a gas detection system.

1207.6.1.2.1 Standby power. Mechanical exhaust ventilation shall be provided with a minimum of 2 hours of standby power in accordance with Section 1203.2.5.

C The intent of this section is where mechanical ventilation is required it must be provided with a minimum of 2 hours of standby power.

Some battery technologies within certain ESS do not continue to generate off gases when the power is out.

1207.6.1.2.2 Installation instructions. Required mechanical exhaust ventilation systems shall be installed in accordance with the manufacturer's installation instructions and the *International Mechanical Code*.

C This section is merely ensuring that ventilation systems are installed as intended by the manufacturer and as required by the IMC for optimal performance.

1207.6.1.2.3 Supervision. Required mechanical exhaust ventilation systems shall be supervised by an *approved* central station, proprietary or remote station service in accordance with NFPA 72, or shall initiate an audible and visible signal at an *approved* constantly attended on-site location.

C Where mechanical ventilation is utilized to keep the concentration of flammable gases, such as hydrogen, from reaching a flammable or explosive limit, some method of ensuring that the ventilation system is operational is necessary. This section provides two options, which include supervision or an audible and visual alarm at an approved location. A security desk that is operated 24 hours a day would be a reasonable location to meet the "constantly attended" requirement.

It should be noted that according to the IEEE 1635, ventilation rates can be very low. Hydrogen is extremely small and will pass through walls and ceilings. Maintaining a concentration of hydrogen in building construction is difficult. One example of a facility exploding due to hydrogen from batteries is the Rancho Cordova, California, data center explosion. The facility was unoccupied and not air conditioned, with batteries on charge with no load. It took months for the hydrogen to build to explosive levels. When treated properly, aqueous batteries do not explode. In many cases, the normal ventilation for human comfort in accordance with the IMC is more than sufficient; however, supervision is required for the ventilation system.

1207.6.1.2.4 Gas detection system. Where required by Section 1207.6.1.2, rooms, areas and walk-in units containing ESS shall be protected by an *approved* continuous *gas detection system* that complies with Section 916 and with the following:

1. The *gas detection system* shall be designed to activate the mechanical ventilation system when the level of *flammable gas* in the room, area or walk-in unit exceeds 25 percent of the LFL.
2. The mechanical ventilation system shall remain on until the *flammable gas* detected is less than 25 percent of the LFL.
3. The *gas detection system* shall be provided with a minimum of 2 hours of standby power in accordance with Section 1203.2.5.
4. Failure of the *gas detection system* shall annunciate a trouble signal at an *approved* central station, proprietary or remote station service in accordance with NFPA 72, or shall initiate an audible and visible trouble signal at an *approved* constantly attended on-site location. (Material based on NFPA 855 2023 Ed.)

C This section states the actions that must occur once the gas detection system activates. Since gas detection is an alternative to continuously running ventilation (see Section 1207.6.1.2), when the gas detection system detects that 25 percent of the LFL has been exceeded, the ventilation system must be activated. Once the flammable gas is below 25 percent, the mechanical ventilation is no longer necessary. Standby power is required for a minimum of 2 hours for such systems in case an event occurs during a power outage. That is consistent with the standby power required by Section 1207.6.1.2.1. Finally, another layer of protection is required by Item 4, which is detection of a failure of the gas detection system.

1207.6.2 Spill control and neutralization. Where required by Table 1207.6 or elsewhere in this code, areas containing free-flowing liquid electrolyte or hazardous materials shall be provided with spill control and neutralization in accordance with this section. (Material based on NFPA 855 2023 Ed.)

C This section is specific to free-flowing liquid electrolyte or other free-flowing liquid hazardous materials. Spill control and neutralization are required by Table 1207.6 for most types of battery technologies. Lithium-ion and nonvented lead-acid, nickel-cadmium and nickel metal hydride batteries would not require this protection since there is little to no risk of liquid spills. Documentation can be provided that demonstrates spill control and neutralization are not necessary for ESS technologies not specifically called out in Table 1207.6.

Batteries that contain a free-flowing liquid electrolyte pose the same containment problems as any other corrosive liquid hazardous materials, but the containment and neutralization provisions in this section are performance based and neither specifically require spill control in the form of containment nor a specific method of neutralization. The quantity of neutralization material required to be available would be greater for less-viscous electrolytes because of their mobility and the rapidity with which they can spread and the potential scope of the spread.

1207.6.2.1 Spill control. Spill control shall be provided to prevent the flow of liquid electrolyte or hazardous materials to adjoining rooms or areas. The method shall be capable of containing a spill from the single largest battery or vessel. (Material based on NFPA 855 2023 Ed.)

C This section simply sets the performance criteria for spill control: stopping the flow from entering adjoining rooms or areas and basing that on the largest single battery.

1207.6.2.2 Neutralization. An *approved* method that is capable of neutralizing spilled liquid electrolyte from the largest battery or vessel to a pH between 5.0 and 9.0 shall be provided. (Material based on NFPA 855 2023 Ed.)

C In addition to spill control where a battery releases liquid electrolyte, the liquid must be neutralized. This requirement is specifically performance based to be flexible for different technologies.

1207.6.2.3 Communications utilities. The requirements of Section 1207.6.2 shall apply only when the aggregate capacity of multiple vessels exceeds 1,000 gallons (3785 L) for lead-acid and nickel-cadmium battery systems operating at less than 50 VAC and 60 VDC that are located at facilities under the exclusive control of communications utilities, and those facilities comply with NFPA 76 in addition to applicable requirements of this code.

C This section recognizes the good safety record and level of proper enforcement through compliance with NFPA 76 for these particular technologies in this application.

1207.6.3 Explosion control. Where required by Table 1207.6 or elsewhere in this code, explosion control complying with Section 911 shall be provided for rooms, areas, ESS cabinets or ESS walk-in units containing electrochemical ESS technologies.

Exceptions: (Material based on NFPA 855 2023 Ed.)

1. Where *approved*, explosion control is permitted to be waived by the *fire code official* based on large-scale fire testing complying with Section 1207.1.7 that demonstrates that *flammable gases* are not liberated from electrochemical ESS cells or modules.
2. Where *approved*, explosion control is permitted to be waived by the *fire code official* based on documentation provided in accordance with Section 104.2.2 that demonstrates that the electrochemical ESS technology to be used does not have the potential to release *flammable gas* concentrations in excess of 25 percent of the LFL anywhere in the room, area, walk-in unit or structure under thermal runaway or other fault conditions.
3. Where *approved*, ESS cabinets that have no debris, shrapnel or enclosure pieces ejected during large-scale fire testing complying with Section 1207.1.5 shall be permitted in lieu of providing explosion control complying with Section 911.
4. Explosion control is not required for lead-acid and nickel-cadmium battery systems less than 50 VAC, 60 VDC in telecommunication facilities under the exclusive control of communications utilities located in building spaces or walk-in units used exclusively for such installations.
5. Explosion control is not required for lead-acid and nickel-cadmium systems used for DC power for control of substations and control or safe shutdown of generating stations under the exclusive control of the electric utility, located in building spaces or walk-in units used exclusively for such installations.
6. Explosion control is not required for lead-acid battery systems in uninterruptable power supplies *listed* and *labeled* in accordance with UL 1778, utilized for standby power applications, and housed in a single cabinet in a single *fire area* in buildings or walk-in units.

C This section addresses a potential hazard associated with explosion. Lithium-ion battery systems and other electrochemical ESS technologies have the potential to rapidly build up potentially explosive atmospheres in the battery or electrochemical ESS room or enclosure under thermal runaway and other conditions, which could result in a catastrophic fire and/or explosion. To protect against these hazards, explosion control in accordance with Section 911 is required for certain battery technologies. This section applies to rooms, areas, ESS cabinets and ESS walk-in units. Note that Table 1207.6 requires explosion control for all technologies except flow control and lead-acid, nickel-cadmium and nickel metal hydride batteries at communications utilities complying with NFPA 76. The exemption for lead-acid, nickel-cadmium and nickel

metal hydride relates to the safety record of these applications specific to communications utilities. There are also several specific exceptions to this section focused on lead acid and nickel cadmium installations.

Exception 1 allows the elimination of explosion control where a large-scale fire test demonstrates that flammable bases are not liberated from the ESS. Exception 2 does not specifically call for a large-scale test, but technical justification is required to demonstrate that a dangerous concentration of flammable gases is not released.

Exception 3 allows the fire code official to omit explosion control where ESS cabinets subjected to large-scale fire testing ejected no debris, shrapnel or enclosure pieces.

Exceptions 4 and 5 align exceptions for telecommunication and utility companies with similar exceptions throughout Section 1207.

Exception 6 omits explosion control for lead-acid battery systems in uninterruptable power supplies listed and labeled in accordance with UL 1778, utilized for standby power applications and housed in a single cabinet in a single fire area in buildings or walk-in units.

1207.6.4 Safety caps. Where required by Table 1207.6 or elsewhere in this code, vented batteries and other ESS shall be provided with flame-arresting safety caps.

C The types of batteries regulated by this section vent hydrogen and oxygen to the atmosphere. There are two types of vented batteries: nonrecombinant and recombinant. Essentially, a nonrecombinant battery is a storage battery in which, under conditions of normal use, hydrogen and oxygen gases created by electrolysis are vented into the air outside of the battery. A recombinant battery is a storage battery in which, under conditions of normal use, most of the hydrogen and oxygen gases created by electrolysis are converted back into water inside the battery instead of venting into the air.

Another term used for nonrecombinant batteries is “flooded” or “vented.” Other terms used to denote recombinant batteries are “valve regulated” or “valve-regulated lead-acid (VRLA).” Flooded batteries vent regularly whereas valve-regulated batteries vent when the internal pressure exceeds the design pressure of their self-resealing vents, which are an integral part of their design. In case of either flooded or VRLA cells, flame-arresting vents are needed to prevent a static spark (or other flame source) outside the battery from propagating to the interior of the battery where oxygen or hydrogen may have accumulated during normal operation of the battery.

Essentially, all ESS technologies except lithium-ion and flow batteries require safety caps. Other technologies not named in Table 1207.6 would need to provide documentation demonstrating that safety caps are not necessary.

1207.6.5 Thermal runaway. Where required by Table 1207.6 or elsewhere in this code, batteries and other ESS shall be provided with a *listed* device or other *approved* method to prevent, detect and minimize the impact of thermal runaway.

C This requirement is applicable only where required by the battery-specific requirements of Table 1207.6. Under certain extreme conditions of high ambient temperature or charging rate, or due to internal partial shorting in cells, VRLA batteries may experience a phenomenon known as “thermal runaway.” This is a condition that occurs in a battery (especially valve-regulated types) where charging energy results in heat generation within the battery greater than the heat dissipated, causing an uncontrolled rise in battery temperature. Under these conditions, a battery may generate excessive heat that can cause failure through cell dryout or, in rare cases, rupture or melting of the battery.

Lithium-ion technologies are also subject to thermal runaway. Most often, damage or a short circuit can lead to runaway. The chemicals that are released during thermal runaway can be toxic and in extreme cases, thermal runaway can cause electrical fires and/or batteries to explode. The ambient air temperature in the battery environment must also be properly maintained. Controlling these factors reduces the potential for thermal runaway.

This section requires either a listed device that can control thermal runaway or other approved methods. There are a variety of ESS types, so a specific method cannot be prescribed to address this hazard. The only battery technology not required to address thermal runaway are flow batteries and vented lead-acid batteries. Lithium-ion batteries are permitted to address thermal runaway through a battery management system in accordance with UL 9540 (see Table 1207.6, Note e).

1207.7 Indoor installations. Indoor ESS installations shall be in accordance with Sections 1207.7.1 through 1207.7.4. (Material based on NFPA 855 2023 Ed.)

C Section 1207.7 and Table 1207.7 address indoor locations and identify two types of indoor installations, dedicated-use installations (typical of utility grid-related facilities) and nondedicated-use installations (typical of ESS in mixed-use buildings or incidental-use areas of occupancies). Protection for each installation is commensurate with the related risk and exposures.

TABLE 1207.7—INDOOR ESS INSTALLATIONS

COMPLIANCE REQUIRED		DEDICATED-USE BUILDINGS ^a	NONDEDICATED-USE BUILDINGS ^b
Feature	Section		
Dwelling units and sleeping units	1207.7.3	NA	Yes
Elevation	1207.5.3	Yes	Yes
Fire suppression systems	1207.5.5	Yes ^c	Yes
Fire-resistance-rated separations	1207.7.4	Yes	Yes
General installation requirements	1207.4	Yes	Yes
Maximum allowable quantities	1207.5.2	No	Yes
Size and separation	1207.5.1	Yes	Yes
Smoke and automatic fire detection ^e	1207.5.4	Yes ^d	Yes
Technology specific protection	1207.6	Yes	Yes
NA = Not Allowed. a. See Section 1207.7.1. b. See Section 1207.7.2. c. Where <i>approved</i> by the fire code official, fire suppression systems are permitted to be omitted in dedicated-use buildings located more than 100 feet (30.5 m) from buildings, lot lines, public ways, stored combustible materials, hazardous materials, high-piled stock and other exposure hazards. d. Where <i>approved</i> by the fire code official, alarm signals are not required to be transmitted to a central station, proprietary or remote station service in accordance with NFPA 72, or a constantly attended location where local fire alarm annunciation is provided and trained personnel are always present. e. Lead-acid and nickel-cadmium battery systems installed in Group U buildings and structures less than 1,500 square feet (139 m ²) under the exclusive control of communications utilities, and operating at less than 50 VAC and 60 VDC in accordance with NFPA 76, are not required to have an <i>approved</i> automatic smoke or fire detection system.			

C Table 1207.7 maps out which safety requirements are applicable to dedicated-use buildings and nondedicated-use buildings. The requirements focus on location and elevation of installations, whether fire suppression is required, how such installations shall be separated (fire resistance and distance), smoke and fire detection requirements and how the technology-specific requirements apply. The main differences between the two types of installations are that dedicated-use buildings are not limited by MAQs and have no specific exposure concerns for dwelling and sleeping units. They are nonoccupied so the life safety risk is much lower. In addition, Note c allows omission of fire suppression if the dedicated-use building is more than 100 feet from occupied buildings since it poses a low life safety hazard. Alarm signals are not required to be transmitted if local alarm annunciation is provided along with constantly present, trained personnel. Dedicated-use buildings, overall, offer less risk to human life as such buildings are occupied only by those running the facility.

1207.7.1 Dedicated-use buildings. For the purpose of Table 1207.7, dedicated-use ESS buildings shall be classified as Group F-1 occupancies and comply with all the following:

1. The building shall only be used for ESS, electrical energy generation and other electrical grid-related operations.
2. Occupants in the rooms and areas containing ESS are limited to personnel that operate, maintain, service, test and repair the ESS and other energy systems.
3. No other occupancy types shall be permitted in the building.
4. Administrative and support personnel shall be permitted in areas within the buildings that do not contain ESS, provided that:
 - 4.1. The areas do not occupy more than 10 percent of the *building area* of the story in which they are located.
 - 4.2. A *means of egress* is provided from the incidental use areas to the *public way* that does not require occupants to traverse through areas containing ESS or other energy system equipment. (Material based on NFPA 855 2023 Ed.)

C Such facilities focus only on ESS, electrical energy generation or other electrical grid-related operations. The only people that may be present are those that operate, maintain, service test or repair such facilities. This section requires that the occupancy classification is F-1. This is also denoted in IBC Section 306.2. Such facilities cannot contain other occupancies. Due to the restrictive use of such buildings, more flexibility is given to quantities of ESS and in certain locations, as addressed in Table 1207.7, omission of fire suppression systems or smoke or fire detection is allowed.

1207.7.2 Nondedicated-use buildings. For the purpose of Table 1207.7, nondedicated-use buildings include all buildings that contain ESS and do not comply with Section 1207.7.1 dedicated-use building requirements. (Material based on NFPA 855 2023 Ed.)

C Nondedicated-use buildings are simply any buildings or facilities that do not comply with Section 1207.7.1. Once other occupancies and associated occupants are introduced, the risk to life safety increases.

1207.7.3 Dwelling units and sleeping units. ESS shall not be installed in *sleeping units* or in *habitable spaces* of *dwelling units*. (Material based on NFPA 855 2023 Ed.)

C This section simply prohibits ESS within sleeping units in R-1 and R-2 occupancies entirely and in portions of dwelling units considered to be habitable spaces, such as a bedroom, due to the various hazards associated with ESS.

1207.7.4 Fire-resistance-rated separations. Rooms and areas containing ESS shall include *fire-resistance-rated* separations as follows:

1. In dedicated-use buildings, rooms and areas containing ESS shall be separated from areas in which administrative and support personnel are located.
2. In nondedicated-use buildings, rooms and areas containing ESS shall be separated from other areas in the building.

Separation shall be provided by 2-hour *fire barriers* constructed in accordance with Section 707 of the *International Building Code* and 2-hour *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, as appropriate. (Material based on NFPA 855 2023 Ed.)

C Essentially, whether dedicated use or nondedicated use, certain aspects of such buildings must be separated by 2-hour fire barriers or 2-hour horizontal separations. The focus is on separating occupied areas from ESS.

1207.8 Outdoor installations. Outdoor installations shall be in accordance with Sections 1207.8.1 through 1207.8.3. *Exterior wall* installations for individual ESS units not exceeding 20 kWh shall be in accordance with Section 1207.8.4. (Material based on NFPA 855 2023 Ed.)

C This section covers two types of outdoor installations, remote outdoor installations (more than 100 feet from exposures) and installations near exposures (100 feet or less, which is more typical of an urban environment). Also note that ESS units are permitted to be installed on exterior walls where they do not exceed 20 kWh. Compliance with Section 1207.8.4 is required. Section 1207.8.4 limits the capacity of such systems and requires each ESS unit to be separated by 3 feet to reduce possible fire exposure to an adjacent system.

TABLE 1207.8—OUTDOOR ESS INSTALLATIONS^a

COMPLIANCE REQUIRED		REMOTE INSTALLATIONS ^a	INSTALLATIONS NEAR EXPOSURES ^b
Feature	Section		
All ESS installations	1207.4	Yes	Yes
Clearance to exposures	1207.8.3	Yes	Yes
Fire suppression systems	1207.5.5	Yes ^c	Yes
Maximum allowable quantities	1207.5.2	No	Yes
Maximum enclosure size	1207.5.6	Yes	Yes
Means of egress separation	1207.5.8	Yes	Yes
Size and separation	1207.5.1	No	Yes ^d
Smoke and automatic fire detection	1207.5.4	Yes	Yes
Technology-specific protection	1207.6	Yes	Yes
Vegetation control	1207.5.7	Yes	Yes
a. See Section 1207.8.1. b. See Section 1207.8.2. c. Where approved by the fire code official, fire suppression systems are permitted to be omitted. d. In outdoor walk-in units, spacing is not required between ESS units and the walls of the enclosure.			

C Table 1207.8 provides the triggers for compliance for outdoor ESS installations. Outdoor installations are classified as either a remote installation or an installation located near exposures. The differentiating aspect is location. Where an installation is more than 100 feet from buildings, lot lines, public ways, stored combustible materials, hazardous materials, high-piled stock and other exposures, it can be treated as remote. All other installations would be dealt with as installations near exposures and are more likely in urban or suburban locations.

Remote installations are excused from some safety features due to the lower impact on life safety. There is no limit on quantities; however, there is a maximum enclosure size. The intent of the maximum enclosure size is to differentiate between indoor storage (building) versus outdoor installation. Section 1207.5.6 corresponds to the largest ISO-type containers used for these installations. Size and separation are also not limited, which means there is no limit on capacity of each ESS or separation from the walls or other ESS.

Notes a and b simply clarify which ESS installations are considered remote and which are not. Note c gives the authority to the fire code official to omit the requirement for fire suppression. This is especially important for very remote installations where the risk is low and the water supply availability or infrastructure to support such systems is not available. Finally, Note d allows an exception to the spacing from the wall for nonremote installations. The maximum size of 50kWh and the 3-foot separation from other groups still applies. As discussed, remote installations have no size or separation limitations for ESS groups.

1207.8.1 Remote outdoor installations. For the purpose of Table 1207.8, remote outdoor installations include ESS located more than 100 feet (30 480 mm) from buildings, lot lines, public ways, stored combustible materials, hazardous materials, high-piled stock and other exposure hazards. (Material based on NFPA 855 2023 Ed.)

C This section establishes what can be considered a remote installation. As discussed in the commentary to Table 1207.8, such installations are not required to address quantities or size and separation requirements. However, many other features such as smoke and automatic fire detection are still required.

1207.8.2 Installations near exposures. For the purpose of Table 1207.8, installations near exposures include all outdoor ESS installations that do not comply with Section 1207.8.1 remote outdoor location requirements. (Material based on NFPA 855 2023 Ed.)

C Essentially, any outdoor installations that cannot be treated as remote must comply with the criteria for those installations 100 feet or closer to exposures.

1207.8.3 Clearance to exposures. ESS located outdoors shall be separated by a minimum of 10 feet (3048 mm) from the following exposures:

1. Lot lines.
2. Public ways.
3. Buildings.
4. Stored combustible materials.
5. Hazardous materials.
6. High-piled stock.
7. Other exposure hazards.

Exceptions: (Material based on NFPA 855 2023 Ed.)

1. Clearances are permitted to be reduced to 3 feet (914 mm) where a 1-hour free-standing *fire barrier* suitable for exterior use and extending 5 feet (1524 mm) above and 5 feet (1524 mm) beyond the physical boundary of the ESS installation is provided to protect the exposure.
2. Clearances to buildings are permitted to be reduced to 3 feet (914 mm) where noncombustible *exterior walls* with no openings or combustible overhangs are provided on the wall adjacent to the ESS and the *fire-resistance rating* of the *exterior wall* is a minimum of 2 hours.
3. Clearances to buildings are permitted to be reduced to 3 feet (914 mm) where a weatherproof enclosure constructed of noncombustible materials is provided over the ESS, and it has been demonstrated that a fire within the enclosure will not ignite combustible materials outside the enclosure based on large-scale fire testing complying with Section 1207.1.7.

C This section primarily applies to installations that are not remote since such exposures are not permitted for remote installations. This section deals with the separation from exposure hazards with the intent of reducing the risk of an ESS failure and any resulting fire or explosion. Likewise, a failure of the ESS should not pose a hazard to surrounding exposures. There are three exceptions that reduce the separation to 3 feet. These options become critical in densely populated areas. The first two focus on the use of fire-resistance-rated construction to separate the ESS from exposures. The distance can be reduced to 3 feet for both. Exception 1 allows a 1-hour rating where a stand-alone barrier is used. Exception 2 allows the use of building construction with no openings but requires a 2-hour rating. Exception 3 reduces the separation to 3 feet where an ESS installation is protected by a weatherproof enclosure of noncombustible construction that demonstrates during a large-scale fire test that the fire will not spread beyond the enclosure.

1207.8.4 Exterior wall installations. ESS shall be permitted to be installed outdoors on *exterior walls* of buildings when all of the following conditions are met:

1. The maximum energy capacity of individual ESS units shall not exceed 20 kWh.
2. The ESS shall comply with applicable requirements in Section 1207.
3. The ESS shall be installed in accordance with the manufacturer's instructions and their listing.
4. Individual ESS units shall be separated from each other by at least 3 feet (914 mm).
5. The ESS shall be separated from doors, windows, operable openings into buildings or HVAC inlets by at least 5 feet (1524 mm).

Exception: Where *approved*, smaller separation distances in Items 4 and 5 shall be permitted based on large-scale fire testing complying with Section 1207.1.7. (Material based on NFPA 855 2023 Ed.)

C Exterior wall installations are also considered outdoor installations. They are permitted only if they comply with the conditions set forth in this section. In addition to general compliance with Section 1207 and the use of properly installed and listed equipment, these ESS units are limited to 20 kWh and must be separated from one another by 3 feet. In terms of exposures to the building they are located near, such installations need to be at least 5 feet from openings, including HVAC inlets. The exception provides the flexibility to conduct large-scale fire testing to allow smaller separation distances.

1207.9 Special installations. Rooftop and open parking garage ESS installations shall comply with Sections 1207.9.1 through 1207.9.6.

C This section deals with two types of installations: rooftop and open parking garages. Table 1207.9 sets out the safety feature compliance requirements. Essentially, the only difference in triggers in this table is that the rooftop installations are to comply with Section 1207.9.5, and open parking garages are to comply with Section 1207.9.6. The table also indicates through Notes a and b the difference between the two types of installations.

TABLE 1207.9—SPECIAL ESS INSTALLATIONS			
COMPLIANCE REQUIRED		ROOFTOPS ^a	OPEN PARKING GARAGES ^b
Feature	Section		
All ESS installations	1207.4	Yes	Yes
Clearance to exposures	1207.9.3	Yes	Yes
Fire suppression systems	1207.9.4	Yes	Yes
Maximum allowable quantities	1207.5.2	Yes	Yes
Maximum enclosure size	1207.5.6	Yes	Yes
Means of egress separation	1207.5.8	Yes	Yes
Open parking garage installations	1207.9.6	No	Yes
Rooftop installations	1207.9.5	Yes	No
Size and separation	1207.5.1	Yes	Yes
Smoke and automatic fire detection	1207.5.4	Yes	Yes
Technology-specific protection	1207.6	Yes	Yes
a. See Section 1207.9.1.			
b. See Section 1207.9.2.			

C This table sets out the compliance features required for such installations. The focus is on clearance to exposures, maximum quantities and configurations, and the need for fire safety systems. Technology-specific requirements reference Section 1207.6, which includes issues such as spill control and neutralization, the need for explosion control and ventilation.

1207.9.1 Rooftop installations. For the purpose of Table 1207.9, rooftop ESS installations are those located on the roofs of buildings. (Material based on NFPA 855 2023 Ed.)

C This section simply scopes what is considered an ESS rooftop installation. Table 1207.9 sets out the features required for compliance.

1207.9.2 Open parking garage installations. For the purpose of Table 1207.9, open parking garage ESS installations are those located in a structure or portion of a structure that complies with Section 406.5 of the *International Building Code*. (Material based on NFPA 855 2023 Ed.)

C This section simply scopes what is considered an ESS open parking garage installation through a reference to IBC Section 406.5. Otherwise, such an installation could likely be considered an indoor installation if the parking garage is considered enclosed, as an example. Table 1207.9 sets out the features required for compliance.

1207.9.3 Clearance to exposures. ESS located on rooftops and in open parking garages shall be separated by a minimum of 10 feet (3048 mm) from the following exposures:

1. Buildings, except the building on which rooftop ESS is mounted.
2. Any portion of the building on which a rooftop system is mounted that is elevated above the rooftop on which the system is installed.
3. Lot lines.
4. Public ways.
5. Stored combustible materials.
6. Locations where motor vehicles can be parked.
7. Hazardous materials.
8. Other exposure hazards.

Exceptions:

1. Clearances are permitted to be reduced to 3 feet (914 mm) where a 1-hour free-standing *fire barrier* suitable for exterior use and extending 5 feet (1524 mm) above and 5 feet (1524 mm) beyond the physical boundary of the ESS installation is provided to protect the exposure.

- Clearances are permitted to be reduced to 3 feet (914 mm) where a weatherproof enclosure constructed of noncombustible materials is provided over the ESS, and it has been demonstrated that a fire within the enclosure will not ignite combustible materials outside the enclosure based on large-scale fire testing complying with Section 1207.1.7. (Material based on NFPA 855 2023 Ed.)

C This section deals with the separation from exposure hazards with the intent of reducing the risk of an ESS failure and any resulting fire or explosion. Likewise, a failure of the ESS should not pose a hazard to exposures such as parked vehicles in the case of an open parking garage. There are two exceptions that reduce the separation to 3 feet. Exception 1 focuses on the use of fire-resistance-rated construction to separate the ESS from exposures. Exception 2 reduces the separation distance requirement where an ESS installation is protected by a weatherproof enclosure of noncombustible construction that demonstrates during a large-scale fire test that the fire will not spread beyond the enclosure.

1207.9.4 Fire suppression systems. ESS located in walk-in units on rooftops or in walk-in units in open parking garages shall be provided with automatic fire suppression systems within the ESS enclosure in accordance with Section 1207.5.5. Areas containing ESS other than walk-in units in open parking structures on levels not open above to the sky shall be provided with an automatic fire suppression system complying with Section 1207.5.5.

Exception: A fire suppression system is not required in open parking garages if large-scale fire testing complying with Section 1207.1.7 is provided that shows that a fire will not impact the exposures in Section 1207.9.3. (Material based on NFPA 855 2023 Ed.)

C The trigger for fire suppression systems is based on being contained within a walk-in unit or being in a location not open to the sky. Compliance with Section 1207.5.5 would not be possible in locations open to the sky. The exception to this section is for open parking structures that, through large-scale testing, do not affect the exposures, as noted in Section 1207.9.3.

1207.9.5 Rooftop installations. ESS and associated equipment that are located on rooftops and not enclosed by building construction shall comply with the following:

- Stairway access to the roof for emergency response and fire department personnel shall be provided either through a bulkhead from the interior of the building or a stairway on the exterior of the building.
- Service walkways at least 5 feet (1524 mm) in width shall be provided for service and emergency personnel from the point of access to the roof to the system.
- ESS and associated equipment shall be located from the edge of the roof a distance equal to at least the height of the system, equipment or component but not less than 5 feet (1524 mm).
- The roofing materials under and within 5 feet (1524 mm) horizontally from an ESS or associated equipment shall be noncombustible or shall have a Class A rating when tested in accordance with ASTM E108 or UL 790.
- A Class I standpipe outlet shall be installed at an *approved* location on the roof level of the building or in the stairway bulkhead at the top level.
- The ESS shall be the minimum of 10 feet (3048 mm) from the fire service access point on the rooftop. (Material based on NFPA 855 2023 Ed.)

C This section is specific to ESS installations on roofs. The requirement focuses on the needs of the fire department. The fire department may be in a position of responding to a fire associated with an ESS installation. Having appropriate features that will assist in this response is necessary. Proper stairway access and the ability to move about the roof is necessary. For further safety, such installations must be moved away from the edge of the roof. This will help firefighters attack from more angles, reduce the risk of falls and decrease the possibility of debris falling from the roof. The access point needs to be of sufficient distance from the ESS installation, and access to a standpipe is necessary in an approved location.

1207.9.6 Open parking garages. ESS and associated equipment that are located in open parking garages shall comply with all of the following:

- ESS shall not be located within 50 feet (15 240 mm) of air inlets for building HVAC systems.
Exception: This distance shall be permitted to be reduced to 25 feet (7620 mm) if the automatic *fire alarm system* monitoring the radiant-energy sensing detectors de-energizes the ventilation system connected to the air intakes upon detection of fire.
- ESS shall not be located within 25 feet (7620 mm) of *exits* leading from the attached building where located on a covered level of the parking structure not directly open to the sky above.
- An *approved* fence with a locked gate or other *approved* barrier shall be provided to keep the general public at least 5 feet (1524 mm) from the outer enclosure of the ESS. (Material based on NFPA 855 2023 Ed.)

C A fire involving an ESS installation in an open parking garage has the ability of exposing occupants both in the parking garage and in surrounding or connected buildings via the HVAC system. Therefore, proper distances from HVAC systems and exits are required. Where HVAC systems are designed to shut down based on the activation of radiant-energy sensing detectors monitoring the ESS, the separation is permitted to be cut in half to 25 feet.

In order to reduce the likelihood of damage to the ESS or injury to occupants, the installation needs to be properly protected with a fence or other approved method.

1207.10 Mobile ESS equipment and operations. Mobile ESS equipment and operations shall comply with Sections 1207.10.1 through 1207.10.7.7. (Material based on NFPA 855 2023 Ed.)

C This section addresses two types of mobile ESS installations/operations: (1) charging and storage of the mobile ESS at its home facility when it is not deployed to an event or facility and (2) deployment of the mobile ESS for temporary energy storage applications, such as providing power at an electric vehicle event. Mobile ESS charging and storage locations are treated the same as a stationary indoor or outdoor installation in accordance with Section 1207.7 or 1207.8, but it can include temporary electrical and fire suppression system connections. This provides an acceptable level of protection based on the exposures at the facility and prevents parties from using an ESS on wheels as a permanent ESS with less than effective protection.

This section also includes requirements for deploying mobile ESS to a facility or event to provide up to 30 days of temporary power (with some exceptions). An operational permit is required for each mobile ESS deployment.

TABLE 1207.10—MOBILE ENERGY STORAGE SYSTEMS (ESS)

COMPLIANCE REQUIRED		DEPLOYMENT ^a
Feature	Section	
All ESS installations	1207.4	Yes ^b
Fire suppression systems	1207.5.5	Yes ^c
Maximum allowable quantities	1207.5.2	Yes
Maximum enclosure size	1207.5.6	Yes
Means of egress separation	1207.5.8	Yes
Size and separation	1207.5.1	Yes ^d
Smoke and automatic fire detection	1207.5.4	Yes ^e
Technology-specific protection	1207.6	Yes
Vegetation control	1207.5.7	Yes

a. See Section 1207.10.2.

b. Mobile operations on wheeled vehicles and trailers shall not be required to comply with Section 1207.4.4 seismic and structural load requirements.

c. Fire suppression system connections to the water supply shall be permitted to use *approved* temporary connections.

d. In walk-in units, spacing is not required between ESS units and the walls of the enclosure.

e. Alarm signals are not required to be transmitted to an *approved* location for mobile ESS deployed 30 days or less.

C This table provides the list of safety features that must be addressed for mobile energy storage systems. These features are similar to other types of installation but also include vegetation control similar to outdoor installations. There are several footnotes. Note a references Section 1207.10.2 for information on what is meant by “deployment.” Note b acknowledges that it is not practical to regulate wheeled vehicles and trailers for seismic and structural requirements. Note c allows the water supply to be a temporary connection. Note d provides an exception for mobile ESS in deployment. Walk-up units do not need to meet the spacing-from-wall requirement of Section 1207.5.1. This is similarly allowed in outdoor installations located within 100 feet of exposures. Note e recognizes the temporary nature of the use and does not require that alarm signals be transmitted to an approved location.

1207.10.1 Charging and storage. For the purpose of Section 1207.10, charging and storage covers the operation where mobile ESS are charged and stored so they are ready for deployment to another site, and where they are charged and stored after a deployment.

Exception: Mobile ESS used to temporarily provide power to lead-acid and nickel-cadmium systems that are used for DC power for control of substations and control or safe shutdown of generating stations under the exclusive control of the electric utility, and located outdoors or in building spaces used exclusively for such installations. (Material based on NFPA 855 2023 Ed.)

C This section describes when such units are not deployed and are being charged and stored. When mobile ESS are charging and stored, they are regulated as applicable to the way in which they are stored. The applicability of the installation types is addressed by Section 1207.10.6. This section has an exception for temporary use at lead-acid and nickel-cadmium systems used for DC power at substations based on the safety record of such applications.

1207.10.2 Deployment. For the purpose of Section 1207.10, deployment covers operations where mobile ESS are located at a site other than the charging and storage site and are being used to provide power.

Exception: Mobile ESS used to temporarily provide power to lead-acid and nickel-cadmium systems that are used for DC power for control of substations and control or safe shutdown of generating stations under the exclusive control of the electric utility, and located outdoors or in building spaces used exclusively for such installations. (Material based on NFPA 855 2023 Ed.)

C When the mobile ESS is in use, it is considered as deployed and all the requirements of Table 1207.10 are applicable versus the requirements found in Sections 1207.7 through 1207.9.

1207.10.3 Permits. Construction and operational permits shall be provided for charging and storage of mobile ESS and operational permits shall be provided for deployment of mobile ESS as required by Section 1207.1.4.

C This section addresses permits for both aspects of ESS used for mobile applications. When mobile ESS are charging and in storage, both construction and operational permits are required. When such systems are deployed, operational permits are required.

1207.10.4 Construction documents. *Construction documents* complying with Section 1207.1.5 shall be provided with the construction permit application for mobile ESS charging and storage locations.

C When mobile ESS are being charged and stored, all the requirements found in Section 1207.1.3 for construction documents must be provided with the construction permit application, which is no different than for other ESS installations.

1207.10.4.1 Deployment documents. The following information shall be provided with the operation permit applications for mobile ESS deployments:

1. Relevant information for the mobile ESS equipment and protection measures in the *construction documents* required by Section 1207.1.5.
2. Location and layout diagram of the area in which the mobile ESS is to be deployed, including a scale diagram of all nearby exposures.
3. Location and content of signage, including no smoking signs.
4. Description of fencing to be provided around the ESS, including locking methods.
5. Details on fire suppression, smoke and automatic fire detection, system monitoring, thermal management, exhaust ventilation and explosion control, if provided.
6. For deployment, the intended duration of operation, including anticipated connection and disconnection times and dates.
7. Location and description of local staging stops during transit to the deployment site. See Section 1207.10.7.5.
8. Description of the temporary wiring, including connection methods, conductor type and size, and circuit overcurrent protection to be provided.
9. Description of how fire suppression system connections to water supplies or extinguishing agents are to be provided.
10. Contact information for personnel who are responsible for maintaining and servicing the equipment, and responding to emergencies as required by Section 1207.1.8.1. (Material based on NFPA 855 2023 Ed.)

C Where the mobile ESS is deployed, additional information is required for the operational permit for such use. Various deployment-specific aspects may affect the safety of the temporary installation, including how fencing and associated locks will be provided, and the location and layout of the area where the mobile ESS will be used. This information is in addition to that required by Section 1207.1.3.

1207.10.5 Approved locations. Locations where mobile ESS are charged, stored and deployed shall be restricted to the locations established on the construction and operational permits. (Material based on NFPA 855 2023 Ed.)

C Mobile ESS charging, storage and deployment locations may vary as long as all locations are those already established through the permitting process. Locations other than those that have been approved could pose safety risks as site and safety features have not been fully reviewed. Additionally, fire department personnel have no way of knowing about possible risks they may encounter when responding to an emergency at a nonpermitted site.

1207.10.6 Charging and storage. Installations where mobile ESS are charged and stored shall be treated as permanent ESS indoor or outdoor installations, and shall comply with the following sections, as applicable:

1. Indoor charging and storage shall comply with Section 1207.7.
2. Outdoor charging and storage shall comply with Section 1207.8.
3. Charging and storage on rooftops and in open parking garages shall comply with Section 1207.9.

Exceptions:

1. Electrical connections shall be permitted to be made using temporary wiring complying with the manufacturer's instructions, the UL 9540 listing and NFPA 70.
2. Fire suppression system connections to the water supply shall be permitted to use *approved* temporary connections. (Material based on NFPA 855 2023 Ed.)

C This section essentially establishes that when mobile ESS are charging or in storage, they should be dealt with using the scoping of Sections 1207.7 through 1207.9. In other words, they are no different than other applications of ESS. Once they are deployed, then Table 1207.10 and special permitting is required.

The exceptions recognize that these types of ESS are intended to be mobile and provide some relief from full compliance with Sections 1207.7 through 1207.9. When not deployed, mobile ESS can use temporary wiring that complies with NFPA 70 and UL 9540 and temporary connections to water supplies for fire protection.

1207.10.7 Deployed mobile ESS requirements. Deployed mobile ESS equipment and operations shall comply with this section and Table 1207.10. (Material based on NFPA 855 2023 Ed.)

C Once mobile ESS are deployed, Sections 1207.10.7.1 through 1207.10.7.7 are applicable. When deployed, there are unique risks that must be addressed that are site and application specific.

1207.10.7.1 Duration. The duration of mobile ESS deployment shall not exceed 30 days.

Exceptions:

1. Mobile ESS deployments that provide power for durations longer than 30 days shall comply with Section 1207.10.6.
2. Mobile ESS deployments shall not exceed 180 days unless additional operational permits are obtained. (Material based on NFPA 855 2023 Ed.)

C The deployment of mobile ESS are critical after natural disasters such as earthquakes and hurricanes but are not intended to replace permanent power sources. This section limits how long such mobile ESS can be used. The main code section limits deployments to 30 days; however, the exceptions address methods of extending that duration for much longer.

Exception 1 is actually a requirement that requires the mobile ESS to be treated as an indoor, outdoor or special application ESS installation. As discussed in the commentary to Section 1207.10.6, there are allowances for wiring and fire protection connections that would apply to mobile ESS that are being stored or charged. Although Section 1207.10.6 allows rooftop and open parking garages for the storing and charging of mobile ESS, Section 1207.10.7.2 would not allow a deployed mobile ESS on a rooftop or in a covered parking garage.

Where mobile ESS have been deployed beyond 30 days under Exception 1, Exception 2 requires additional operational permits for that deployment to exceed 180 days.

1207.10.7.2 Restricted locations. Deployed mobile ESS operations shall not be located indoors, in covered parking garages, on rooftops, below grade or under building overhangs. (Material based on NFPA 855 2023 Ed.)

C This section provides several locations where mobile ESS cannot be deployed. Deployment at these locations creates potential exposure hazards to buildings, occupants and firefighters. These installations, though well regulated, are not as permanent as other ESS and are at a higher risk of being damaged and possibly leading to fire and explosion hazards.

1207.10.7.3 Clearance to exposures. Deployed mobile ESS shall be separated by a minimum of 10 feet (3048 mm) from the following exposures:

1. Public ways.
2. Buildings.
3. Stored combustible materials.
4. Hazardous materials.
5. High-piled storage.
6. Other exposure hazards.

Deployed mobile ESS shall be separated by a minimum of 50 feet (15 240 mm) from public seating areas and from tents, canopies and membrane structures with an *occupant load* of 30 or more. (Material based on NFPA 855 2023 Ed.)

C This section uses separation to reduce risk from exposure hazards of a mobile ESS failure and resulting fire or explosion. Likewise, a failure of the ESS should not pose a hazard to surrounding exposures. This section is similar to Section 1207.8.3 for outdoor installations where a minimum of 10 feet is required to exposures. A key difference is mobile ESS installations can be used for many types of temporary applications. The last sentence addresses a minimum separation distance of 50 feet for public seating as well as tents and canopies with occupant loads over 30.

1207.10.7.4 Electrical connections. Electrical connections shall be made in accordance with the manufacturer's instructions and the UL 9540 listing. Temporary wiring for electrical power connections shall comply with NFPA 70. Fixed electrical wiring shall not be provided. (Material based on NFPA 855 2023 Ed.)

C This section ensures that manufacturer's instructions are used and that temporary wiring complies with NFPA 70. Deployed mobile ESS are not intended for fixed use so they should not have fixed electrical wiring.

1207.10.7.5 Local staging. Mobile ESS in transit from the charging and storage location to the deployment location and back shall not be parked within 100 feet (30 480 mm) of an occupied building for more than 1 hour during transit, unless specifically *approved* by the *fire code official* when the permit is issued. (Material based on NFPA 855 2023 Ed.)

C Mobile ESS in transit are at risk due to the energy contained. To avoid the risk to life safety of occupants in surrounding buildings, this section requires a 100-foot separation where an ESS will be located for a longer period of time. A similar concept is used for other hazardous materials regulated by this code, such as for LP-gas. In addition, it would be difficult to provide an operational permit each time a mobile ESS was in transit.

1207.10.7.6 Fencing. An *approved* fence with a locked gate or other *approved* barrier shall be provided to keep the general public at least 5 feet (1524 mm) from the outer enclosure of a deployed mobile ESS. (Material based on NFPA 855 2023 Ed.)

C This section protects the ESS from damage and provides some separation to protect the general public from exposure to the ESS. Outdoor installations are more subject to vandalism and impact.

1207.10.7.7 Smoking. Smoking shall be prohibited within 10 feet (3048 mm) of mobile ESS. Signs shall be posted in accordance with Section 310.

C Regulating smoking near a deployed mobile ESS installation is key to reducing the likelihood of fire or explosion hazards.

1207.11 ESS in Group R-3 and R-4 occupancies. ESS in Group R-3 and R-4 occupancies shall be in accordance with Sections 1207.11.1 through 1207.11.9.

Exceptions:

1. ESS *listed* and *labeled* in accordance with UL 9540 and marked “For use in residential dwelling units,” where installed in accordance with the manufacturer’s instructions and NFPA 70.
2. ESS rated less than 1 kWh (3.6 megajoules).

C An increased number of ESS utilizing stationary storage batteries are appearing on the market to help meet society’s growing energy demands. There are several driving factors, but a move to NetZero and energy independence are likely the biggest. This section does not mandate that ESS be provided but instead provides safety requirements that need to be applied if such systems are to be installed. It should be noted that Section 1207.11 is very similar to IRC Section R330 for one- and two-family dwellings.

The definition of “Energy storage system” in Chapter 2 provides the code user with information on battery storage systems.

Exception 1 acknowledges installations that are listed and labelled to UL 9540 that are specifically marked for residential dwelling units. They are more specially tested and listed for this purpose.

Exception 2 eliminates an enforcement burden by exempting low energy capacity ESS due to their lack of hazard. The energy capacity of 1 kWh is slightly greater than two 12V, 40 Ah batteries. This exempts common household standby power systems for tools, alarm systems and other appliances from having to comply with this section

1207.11.1 Equipment listings. ESS shall be *listed* and *labeled* in accordance with UL 9540.

C This section provides for the basic listing requirements of UL 9540. This listing standard provides construction and performance requirements for investigating and listing stationary storage battery systems and is required by Section 1207 generally. This standard evaluates the ability of the ESS to operate under both normal operating conditions and certain fault conditions.

1207.11.2 Installation. ESS shall be installed in accordance with the manufacturer’s instructions and their listing. (Material based on NFPA 855 2023 Ed.)

C This section provides the enabling language to require that ESS are installed in accordance with the manufacturer’s instructions and the listing.

1207.11.2.1 Spacing. Individual ESS units shall be separated from each other by at least 3 feet (914 mm) except where smaller separation distances are documented to be adequate based on large-scale fire testing complying with Section 1207.1.7.

C Regardless of whether the ESS complies with the listing and manufacturer’s instructions, this section requires minimal separation of units to reduce the risk of a larger failure if one should occur. There is an allowance provided to reduce separations where large-scale fire testing demonstrates that the fire will not propagate between units. IRC Section R330 approaches spacing slightly differently and depends on the ESS listing, which will include large-scale fire testing when required and the manufacturer’s instructions. This type of testing could occur during the listing process.

1207.11.3 Location. ESS shall be installed only in the following locations:

1. Detached garages and detached accessory structures.
2. Attached garages separated from the *dwelling unit* living space and *sleeping units* in accordance with Section 406.3.2 of the *International Building Code*.
3. Outdoors or on the exterior side of exterior walls located a minimum of 3 feet (914 mm) from doors and windows directly entering the *dwelling unit*.
4. Enclosed utility closets, *basements*, and storage or utility spaces within *dwelling units* and *sleeping units* with finished or noncombustible walls and ceilings. Walls and ceilings of unfinished wood-framed construction shall be provided with not less than 5/8-inch Type X gypsum wallboard.

ESS shall not be installed in sleeping rooms, or in closets or spaces opening directly into sleeping rooms. (Material based on NFPA 855 2023 Ed.)

C This section sets out specific allowable locations for ESS installations that are intended to reduce the risk to occupants where they are most at risk in their homes. These locations are primarily exterior or within attached or detached garages. Item 4 is focused on locations within the sleeping units and dwelling units where ESS must be located and how they are to be separated. This section, along with IRC Section R330.4, specifically prohibits such units in sleeping rooms and spaces that are directly associated with such rooms. People are most at risk where they sleep.

1207.11.4 Energy ratings. Individual ESS units shall have a maximum rating of 20 kWh. The aggregate rating of the ESS shall not exceed:

1. 40 kWh within utility closets, *basements*, and storage or utility spaces.
2. 80 kWh in attached or detached garages and detached accessory structures.
3. 80 kWh on *exterior walls*.
4. 80 kWh outdoors on the ground.

ESS installations exceeding the permitted individual or aggregate ratings shall be installed in accordance with Sections 1207.1 through 1207.9. (Material based on NFPA 855 2023 Ed.)

C This section sets limitations on the maximum energy rating of individual units as well as an aggregate energy rating for specific installation locations. The size of an event is directly correlated to the amount of energy stored. Where more energy rating is desired, the installation is required to comply with Section 1207 more generally.

1207.11.5 Electrical installation. ESS shall be installed in accordance with NFPA 70. Inverters shall be *listed* and *labeled* in accordance with UL 1741 or provided as part of the UL 9540 listing. Systems connected to the utility grid shall use inverters *listed* for utility interaction. (Material based on NFPA 855 2023 Ed.)

C Compliance with NFPA 70 is crucial. Additionally, inverters that are often associated with energy storage and PV installation to convert DC power to AC power must be listed and labeled. This section requires either listing with UL 1741 or UL 9540 where the inverters are part of the listing. In some cases, systems feed energy back to the grid, so the inverters need to be listed for that application as well.

1207.11.6 Fire detection. ESS installed in Group R-3 and R-4 occupancies shall comply with the following:

1. Rooms and areas within *dwelling units*, *sleeping units*, *basements* and attached garages in which ESS are installed shall be protected by smoke alarms in accordance with Section 907.2.11.
2. A *listed* heat alarm shall be installed in locations where smoke alarms cannot be installed based on their listing.

C For early warning of an event, this section requires that an interconnected smoke alarm system with a smoke alarm be installed in the room or space where the ESS is located. If the space is not conducive to the installation of a smoke alarm, a listed heat alarm can be installed and interconnected to the smoke alarm system so that the occupants will be alerted to activation. Smoke alarms or heat alarms, where applicable, are generally going to be required in most locations listed in this section, as required in Section 907.2.11. However, additional detectors may be required based on the specific location of the ESS installation. Section 907.2.11.7 would permit the installation of a fire alarm system utilizing UL 268-listed smoke detectors installed in accordance with Section 907 and NFPA 72.

1207.11.7 Protection from impact. ESS installed in a location subject to vehicle damage in accordance with Section 1207.11.7.1 or 1207.11.7.2 shall be provided with impact protection in accordance with Section 1207.11.7.3.

C To reduce the risk of a failure of an ESS unit due to impact, impact protection is required. Impact could create a situation where thermal runaway occurs. The need for vehicle impact protection is established separately for garages and all other locations within Sections 1207.11.7.1 and 1207.11.7.2. Once the need for impact protection is established, three options are provided in Section 1207.11.7.3.

1207.11.7.1 Garages. Where an ESS is installed in the normal driving path of vehicle travel within a garage, impact protection complying with Section 1207.11.3 shall be provided. The normal driving path is a space between the garage vehicle opening and the interior face of the back wall to a height of 48 inches (1219 mm) above the finished floor. The width of the normal driving path shall be equal to the width of the garage door opening. Impact protection shall also be provided for an ESS installed at either of the following locations (see Figure 1207.11.7.1):

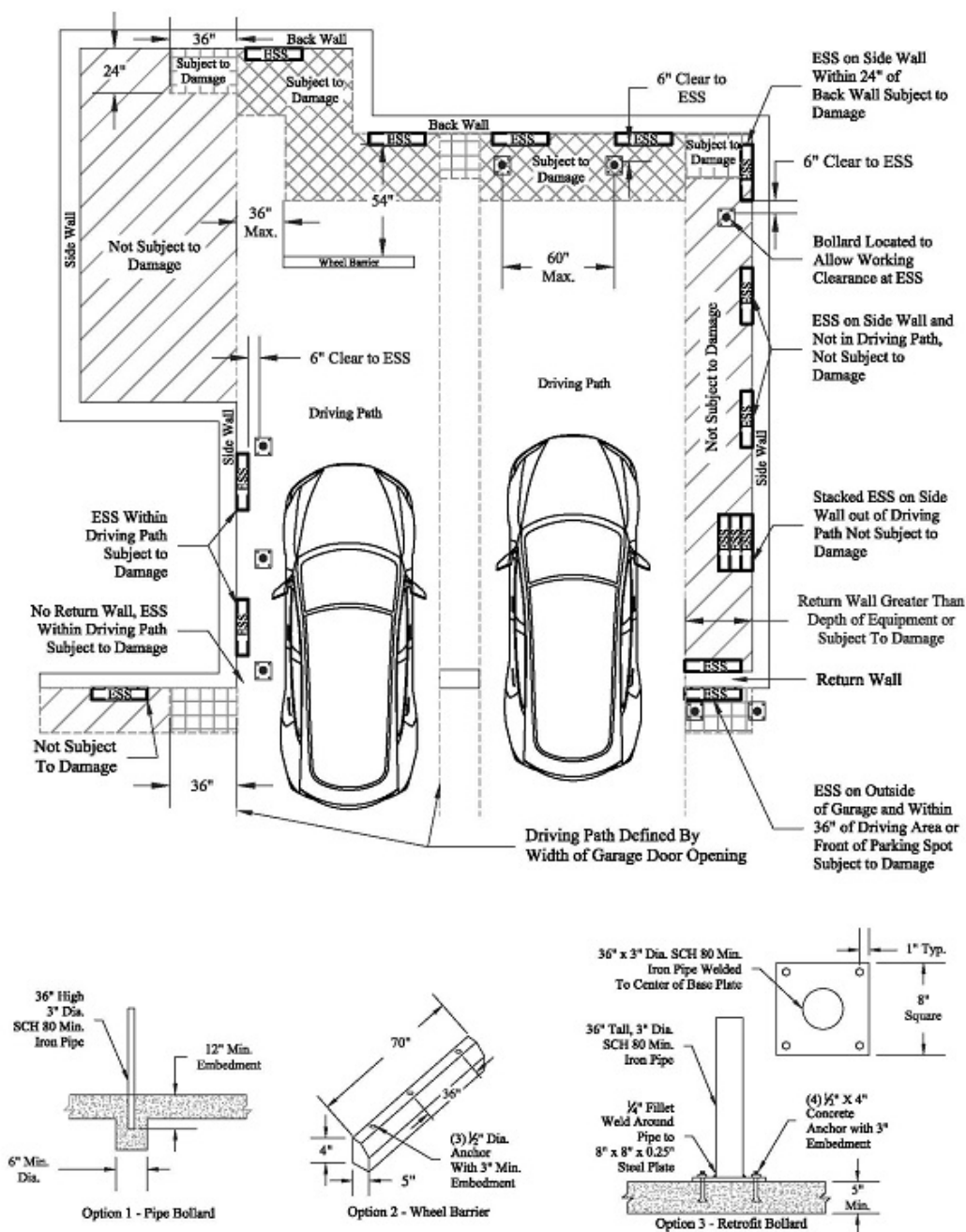
1. On the interior face of the back wall and located within 36 inches (914 mm) to the left or to the right of the normal driving path.

2. On the interior face of a side wall and located within 24 inches (610 mm) of the back wall and 36 inches (914 mm) of the normal driving path.

Exception: Where the clear height of the vehicle garage opening is 7 feet 6 inches (2286 mm) or less, ESS installed not less than 36 inches (914 mm) above the finished floor are not subject to vehicle impact protection requirements.

C This section requires that impact protection be provided where the ESS is in the normal driving path of the vehicle travel. The section establishes specific criteria as to what constitutes a “normal driving path,” which is essentially any space within the width of the door opening up to 48 inches above the finished floor. Items 1 and 2 detail impact protection on the back and side walls, respectively, in relation to their proximity to the normal driving path. Figure 1207.11.7.1 provides more detailed guidance as to which areas need protection and which do not. The various options are important where protection installation is complicated or prevented by post or pretensioned slab construction.

FIGURE 1207.11.7.1—ESS VEHICLE IMPACT PROTECTION



C See the commentary to Section 1207.11.7.1.

1207.11.7.2 Other locations subject to vehicle impact. Where an ESS is installed in a location other than as defined in Section 1207.11.7.1 and is subject to vehicle damage, impact protection shall be provided in accordance with Section 1207.11.7.3.

C This section addresses locations aside from garages. An example may be an exterior wall installation located next to a home's driveway. This section is a bit more subjective but, essentially, if the ESS is vulnerable to impact based on placement, the impact protection requirements provided in Section 1207.11.7.3 must be applied.

1207.11.7.3 Impact protection options. Where ESS is required to be protected from impact in accordance with Section 1207.11.7.1 or 1207.11.7.2, such protection shall comply with one of the following:

1. Bollards constructed in accordance with one of the following:
 - 1.1. Minimum 48 inches (1219 mm) in length by 3 inches (76 mm) in diameter Schedule 80 steel pipe embedded in a concrete pier not less than 12 inches (304 mm) deep and 6 inches (152 mm) in diameter, with at least 36 inches (914 mm) of pipe exposed, filled with concrete and spaced at a maximum interval of 5 feet (1524 mm). Each bollard shall be located not less than 6 inches (152 mm) from an ESS.
 - 1.2. Minimum 36 inches (914 mm) in height by 3 inches (76 mm) in diameter Schedule 80 steel pipe fully welded to a minimum 8 inches (203 mm) by ¼-inch (6.4 mm) thick steel plate and bolted to a concrete floor by means of four ½-inch (13 mm) concrete anchors with 3-inch (76 mm) minimum embedment. Spacing shall be not greater than 60 inches (1524 mm), and each bollard shall be located not less than 6 inches (152 mm) from the ESS.
 - 1.3. Premanufactured steel pipe bollards shall be filled with concrete and anchored in accordance with the manufacturer's installation instructions, with spacing not greater than 60 inches (1524 mm). Each bollard shall be located not less than 6 inches (152 mm) from the ESS.
2. Wheel barriers constructed in accordance with one of the following:
 - 2.1. Four inches (102 mm) in height by 5 inches (127 mm) in width by 70 inches (1778 mm) in length wheel barrier made of concrete or polymer, anchored to the concrete floor not less than every 36 inches (914 mm) and located not less than 54 inches (1372 mm) from the ESS. Minimum 3½-inch (89 mm) diameter concrete anchors with 3-inch (76 mm) embedment per barrier shall be used. Spacing between barriers shall be not greater than 36 inches (914 mm).
 - 2.2. Premanufactured wheel barriers shall be anchored in accordance with the manufacturer's installation instructions.
3. *Approved* method designed to resist a 2,000-pound-force (8896 N) impact in the direction of travel at 24 inches (610 mm) above grade.

C This section provides three options with detailed associated criteria. The first two options are very prescriptive and focus on the installation of either bollards or wheel barriers. The third option is performance based and simply provides the performance criteria of such impact being able to withstand a certain force at a prescribed height.

1207.11.8 Ventilation. Indoor installations of ESS that include batteries that produce hydrogen or other *flammable gases* during charging shall be provided with exhaust ventilation in accordance with Section 304.5 of the *International Mechanical Code*. (Material based on NFPA 855 2023 Ed.)

C This section requires ventilation be provided for indoor installations of ESS technologies, such as those including lead-acid batteries that are capable of producing hydrogen gas during charging.

1207.11.9 Electric vehicle use. The temporary use of an *owner* or occupant's electric-powered vehicle to power a *dwelling unit* or *sleeping unit* while parked in an attached or detached garage or outdoors shall comply with the vehicle manufacturer's instructions and NFPA 70. (Material based on NFPA 855 2023 Ed.)

C This section provides for the temporary use of an electric vehicle as an ESS to power the dwelling, provided that it is done in compliance with NFPA 70 and the manufacturer's instructions. Requiring compliance with the manufacturer's instructions ensures that only electric vehicles designed and manufactured for use as an ESS are utilized compared to someone adding nonapproved electrical connections to an electric vehicle that is not designed for this purpose.

Bibliography

The following resource materials were used in the preparation of the commentary for this chapter of the code:

2022 *Annual Solar Photovoltaic Module Shipments* Report (US Energy Information Administration, July 2023), https://www.eia.gov/renewable/annual/solar_photo/pdf/pv_full_2022.pdf

IEEE 1035-2012/ASHRAE 21, *Guide for the Ventilation and Thermal Management of Batteries for Stationary Applications*. Atlanta, GA: ASHRAE, 2012.

Hnatov, Matthew V., *Incidents, Deaths, and In-Depth Investigations Associated with Non-Fire Carbon Monoxide from Engine-Driven Generators and Other Engine-Driven Tools, 2004–2014* (US Consumer Product Safety Commission, 2015). <https://www.cpsc.gov/s3fs-public/pdfs/GeneratorsandOEDTFatalities2015.pdf>